QMX+ Panadapter — User Guide (v1.8.1)

By Steffen Lav (OZ1LAV). Generated from the project README — for the live page, issues, and releases see https://github.com/SteffenLav/qmx-panadapter

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The full documentation also lives online at **tab5.lav.dk** — the user guide, quick-start, and reference as searchable, cross-linked web pages, kept in sync with every release.

Features

Everything below is in the firmware **today**. Nothing here needs a PC, and only the items marked *(needs WiFi)* need a network.

Real-time panadapter — Spectrum and waterfall across a 48 kHz window centred on the QMX VFO, at 30 Hz, with 12 kHz IF-offset compensation. Tap or drag to tune (mode-aware snapping), pinch to zoom ×1–×24 with a true zoom-FFT, one-finger drag to pan and retune. Adaptive per-bin noise floor, flat-spectrum mode, adjustable waterfall black level and contrast, selectable FFT window, and a graphical S-meter calibrated in dBm. A colour band-plan strip tracks the VFO (IARU region 1/2/3, auto-selected from your grid) and can be dragged to scrub across the band.

QMX CAT control — Live frequency, mode (USB/LSB/CW/DiGi, plus AM on QMX firmware 1.04+), SSB filter bandwidth, CW passband, passband overlay, TX power and SWR readout, QMX volume in decibels matching the radio's own display, and an Antenna Tune button on 1.04+. Round-trip CAT latency under 50 ms. Band presets with per-band frequency recall, and 32 memory channels in a 4×8 grid holding any frequency and mode.

FT8 and FT4, receive — Continuous on-device decoding in the same view as the panadapter: FT8 (15 s slots) and FT4 (7.5 s slots). The decode list shows callsign, country, signal report, slot-timing offset (DT), audio tone (HZ), distance and bearing, with the station you are working held at the top. Include/exclude filters match anything in the message text, worked-before stations can be excluded band-by-band, and a pileup tracker collects everyone calling you.

FT8 and FT4, transmit — Tap a station and the correct *next* message is sent (WSJT-X double-click style); or call CQ from one of three editable presets, with an optional stop-after-N-calls limit. Full automatic exchange through to `73` and an ADIF entry, a resend if your partner never heard your final, a polite hold for a station already working someone else, grey-listing for stations that never answer, and an optional unattended auto-answer robot. Your TX tone is a permanent on-screen button with a live occupancy strip, drag-to-pick tone selection, **TX Hold**, and a cycling **TXCQ ANY / EVEN / ODD** time-window choice. ARRL Field Day exchange mode included.

QSO logging and upload — Every contact is written to an ADIF log on the device, with timestamp, callsign, grid, frequency, band, mode, both signal reports and distance (never a report that was not actually exchanged). Read the log on the Tab5 or in the browser, delete single records or all of them, and upload *(needs WiFi)* to **QRZ Logbook**, **eQSL** and **ARRL LoTW** — LoTW QSOs are signed on the device itself with your own callsign certificate.

Live spots on the spectrum *(needs WiFi)* — **POTA** park activations, and optionally **RBN** CW skimmer spots and **DX cluster** spots, drawn onto the trace at the frequency the station is actually using. The cluster is where phone activity comes from — skimmers are machines, and no machine recognises a callsign spoken into a microphone. One station is one entry, however many sources report it. Grey means you have already worked them on that band. Press and drag to pick one, lift to tune it *with the right mode*. Spots fade with age and are gone after 30 minutes; corner counts take you to the ones just off-screen.

Knowing you are getting out *(needs WiFi)* — **Who is hearing me** asks PSK Reporter which receivers have copied *your* call, with distance, bearing and the report they gave you. **SWR protection** cuts a transmission short and latches the transmitter off above your chosen limit, because an FT8 burst is thirteen seconds of key-down into whatever the antenna is doing.

Activation logging — Start a **POTA** or **SOTA** activation and every contact is stamped with the reference as it is logged, counted against the threshold, with chases tagged from spots already seen and a single-reference ADIF export for uploading.

Bluetooth mouse — A pointer that works *while the QMX stays plugged in*, which a USB mouse cannot do on this hardware. Pair once; it reconnects by itself. Move, click, and a wheel that scrolls whatever is under the pointer. **The mouse must be Bluetooth 4.0 or later (Bluetooth Low Energy).** The Tab5's Bluetooth comes from a co-processor with no Bluetooth Classic radio, so an older Classic mouse cannot work and no firmware change can help — it never appears to the Tab5 at all. Most mice sold since about 2014 qualify; a dual-mode mouse works on its Low Energy channel.

PSK Reporter spotting *(needs WiFi)* — The stations you decode are reported to PSK Reporter the way WSJT-X does, so you appear on the map as a monitoring station. **On by default**; sends your call and grid plus each decoded station's call, grid, frequency, report and mode, batched at most once every five minutes — over the internet only, **never on the air**. One checkbox turns it off, and it is inert until your callsign and grid are set, and in simulation mode.

Web UI *(needs WiFi)* — The whole panadapter in any browser on the LAN: live spectrum and waterfall at ~10 fps, click or drag to tune, mouse-wheel pan and tune, band/mode/bandwidth/zoom control, a graphical S-meter and the whole-band plan strip. In FT8/FT4 mode it shows a live TX status banner and a **Call CQ** button instead of the stream. Also: the QSO log as a sortable table, config download/upload as an editable text file, a microSD file browser, screenshots, and the diagnostic log.

Built-in user manual — This entire guide is compiled into the firmware, so it opens instantly with no WiFi, no card and no download, and can never describe a different version than the one you are running. It opens at the chapter for the screen you are on, warning banners are tappable, and a **Need guidance?** panel lets you pick your symptom in plain words. See Getting help.

Time, with or without a network — SNTP when WiFi is up *(needs WiFi)*, the Tab5's own supercap-backed RTC across power-off, the QMX's clock as an offline fallback, GPS detected and phase-locked automatically if your QMX has one, and a manual set-and-sync panel. FT8/FT4 timing also self-corrects from the decoded band consensus when offline.

microSD station backup — Insert a card (a plain FAT32 32 GB card is ideal) **before switching on** and your whole station is mirrored to `/qmx-panadapter/`: the ADIF log, a full config export, your LoTW certificate and key, the diagnostic log and a self-describing `README.txt`. Continuous with WiFi off (green **SD** dot); one complete backup per start-up with WiFi on (yellow dot), because the card and the WiFi co-processor share a bus. *(The card holds credentials — WiFi password, QRZ/eQSL logins, LoTW private key — so keep it physically secure.)*

Diagnostics — An always-on diagnostic log, nothing to enable: 5 MB in RAM, a rolling copy in flash that survives a power cut, and a full mirror to microSD if a card is in. Downloadable from the browser or over USB serial. Bug reports become answerable.

Practice mode — A simulator with phantom stations that call CQ and reply through the real encode/decode pipeline, so you can rehearse a full QSO with **no radio connected** — and with a hard interlock that never keys a QMX even if one is attached.

Touch, keyboard, mouse — Edge-swipe navigation with breathing grip handles, an on-screen keyboard, optional support for the M5Stack Tab5 70-key snap-on keyboard, and USB mouse support. *(The mouse and the QMX cannot share the single USB host port — the ESP32-P4's USB stack lacks the Transaction Translator a hub would need — so the mouse is for setup, log review and reading the manual with the radio unplugged.)*

Built for the field — WiFi is entirely optional; battery percentage and voltage with a charge limit for battery care; display sleep and a 180° flip for awkward mounting; config backup and restore as a text file; and a settings reset that does not need a reflash.

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1. Quick Guide

Step 0 — Check your QMX firmware first

Power the QMX on by itself and read the firmware version off its own display at boot. You need **1.03.002 or newer**. If yours is older, update the QMX *before* connecting the Tab5 — everything that follows depends on it. This takes 10 seconds to verify and saves hours of debugging. Both **1.03.002** and the **1.04 betas** work with the panadapter (1.04.004 is what this is developed against day to day); on 1.04 the Tab5 additionally offers **AM mode** and an **Antenna Tune** button (SWR tune with a live power/SWR readout) — both stay hidden on 1.03.002, so there's no downside either way.

Step 1 — Flash the Tab5 firmware

Use the one-click flasher in `tools/QMX-Panadapter flasher/` (also attached to each release):

1. Plug the Tab5 into your computer with a **USB-C data cable** — charge-only cables will not work.
2. Run the flasher:
   - **Windows** — double-click `flash.bat` (fetches esptool by itself on first run; nothing to install)
   - **macOS** — double-click `flash.command` (needs esptool once — `brew install esptool` recommended; `pip3 install esptool` often fails on recent macOS with an "externally-managed-environment" error). If macOS blocks the script, right-click → **Open**, or run `bash flash.command` in Terminal (no `chmod` needed).
   - **Linux** — `bash flash.command` (needs `pip3 install esptool`)
3. The flasher asks **normal or clean** (see below) — just press **Enter** for a normal flash.
4. Wait for `SUCCESS`. The Tab5 restarts on the new firmware.

The firmware is already in the zip — `bootloader.bin`, `partition-table.bin` and `qmx_panadapter.bin` sit next to the script, and those are what get flashed. Nothing is downloaded except `esptool` itself, once, if your computer doesn't already have it. So after the first successful run the flasher works with no internet at all.

Normal vs clean flash. Just before flashing, the flasher asks:

> *Type E for a CLEAN/ERASE flash, or just Enter for a normal flash*

- **Normal (press Enter)** — updates the firmware and **keeps** all your saved settings (WiFi, callsign, grid, memory channels, ADIF log). This is what you want almost every time.
- **Clean (type E)** — wipes the whole chip first, so **every saved setting is permanently erased**: WiFi name *and* password, callsign/grid, all memory channels, and the logged QSOs. Use it only if something is stuck or corrupted (e.g. WiFi refuses to turn on no matter what). Back up first with **Config ↓** (below) so you can restore in seconds afterwards.

Building from source? See Build from source.

Step 2 — Connect the cables

You need **two** USB connections, and the cable to the QMX is the one people get wrong:

Connection	Port on Tab5	Cable	Carries
Tab5 → QMX	**USB-A** (host)	**USB-A → USB-C, full data cable**	I/Q audio (UAC) + CAT (CDC-ACM)
Tab5 → power	**USB-C**	Any USB-C power cable (5 V / 2 A+)	Power

> **The #1 failure is a charge-only cable.** Many USB-C cables — especially thin ones bundled with phones and chargers — carry power only, no data lines. If you use one between the Tab5's USB-A port and the QMX, the Tab5 powers the QMX but sees no audio and no CAT: the spectrum stays flat and the top bar shows `Band: ---`. Use a cable you know does data (one that works for a USB stick or phone file transfer). When in doubt, swap the cable first.

Power-on order matters. Turn the **Tab5 on first** and let it finish loading, then turn the **QMX on**. Within a few seconds the top bar should populate Band / Mode / BW and the spectrum should come alive.

Once flashed you can power the Tab5 from any 5 V/2 A USB-C source or the internal battery — the laptop is only needed for flashing. For diagnostics the Tab5 also outputs a serial log over the USB-C data connection (useful if WiFi is not available), but for most users the built-in **diagnostic log** — always on, nothing to enable — is the easier path — see Step 7.

Step 3 — Find your way around

The whole app is driven by **one-finger swipes from the screen edges** and **taps on the top bar**. The settings drawer is always one swipe away — no hunting for hidden menus.

Gesture	From	Does
Swipe →	Left edge	Toggle Panadapter ↔ FT8 screen
Swipe ←	Right edge	Open settings drawer
Swipe ↑	Bottom edge	Open memory channel picker
Tap or swipe ↓	Any top-bar item	Open that item's selector

Slim "breathing" grip handles on each edge show where to swipe — faint enough not to clutter the display, visible if you look for them.

The top bar reads **Band · Mode · BW · Freq · Signal · Zoom** left to right. Tap any item to open a selector — **Freq** opens the frequency keypad; **Mode** switches USB/LSB/CW/DiGi; **BW** selects SSB filter width (USB/LSB only); **Band** jumps to a configured band; **Zoom** selects a zoom preset.

See the full gesture table in the Reference section.

Step 3b — Choose FT8 or FT4 mode

FT8 is available on all HF bands; FT4 is available on select bands with published frequencies (20 m, 30 m, 40 m, etc.). To switch between them:

1. **In Panadapter mode:** Tap the **Mode** selector in the top bar, then choose **DiGi** (the digital mode group).
2. **In FT8 screen:** Tap the **Preset** frequency button to open the band/mode picker — you'll see **two columns**: FT8 (gold header) on the left, FT4 (cyan header) on the right. Tap any frequency in the FT4 column to switch modes.

The decode list, TX controls, and slot timing (countdown bar, which-parity-next) all adapt automatically to the active mode. Switching modes clears stale decodes from the previous mode so you don't work old QSOs.

When to use each mode:

- **FT8** — Most populated band in SOTA/POTA; longer decode window gives more time to prepare replies on slow links or weak signals.

- **FT4** — Faster QSO cycle for busy bands (Field Day, major contests, 20 m pile-ups) and operators who prefer shorter waits between exchanges.

Step 4 — Fill in your settings

On first boot the Tab5 opens a setup prompt automatically — it will ask for your callsign, grid square, and WiFi credentials before you reach the main screen. You can skip any field and fill it in later via the settings drawer.

To open the settings drawer at any time: swipe in from the right edge, or tap the right grip handle.

- **WiFi** — enter your SSID and password. Recommended for everyone: you get accurate UTC time (needed for FT8 slot timing), the browser web UI, and remote diagnostics. The **WiFi initiated** checkbox below the WiFi section is an on/off master switch — uncheck it for POTA or field use with no network.
- **Identity** (callsign + grid square) — *only required for FT8/FT4 transmit*. Enter your callsign and Maidenhead grid (4 or 6 characters, e.g. `JO45` or `JO45ab`). The grid also drives the distance and bearing columns in the FT8 decode list.

Everything else — dB range, smoothing, colour map, brightness, IQ balance — has sane defaults. Adjust if you want to, but you don't need to on first use.

Step 5 — Using the panadapter

Switch the QMX to any band and watch it come alive. This is the foundation of the device regardless of which modes you operate.

Tap or drag to tune. Touch anywhere on the spectrum or waterfall to place the cyan cursor. Drag and it snaps to a mode-aware grid — 10 Hz steps in CW, 250 Hz in SSB, 500 Hz for FT8/FT4 — so you can land precisely on a signal before lifting your finger. Lift, and the QMX retunes.

Swipe fast to pan instead. A quick horizontal swipe (rather than a held drag) slides the whole view left or right and retunes to wherever you let go — the fastest way to slide along a band. See Touch-to-tune for exactly how the two gestures are told apart.

Or move precisely with the band-plan slider. For a controlled move rather than a quick swipe, drag the framed "visible window" on the band-plan strip (along the bottom) to exactly where you want to be on the band — the spectrum, waterfall, and VFO all follow it, so you can place yourself precisely on a segment before you even tap a signal.

Zoom in on a crowded band. Pinch with two fingers to zoom up to ×24. On a CW band at ×8 or higher you can resolve individual signals a few hundred Hz apart, read the spacing between callers, and pick your target before you tune. The frequency axis scales with you, down to Hz precision at high zoom. Double-tap anywhere to reset to the full 48 kHz view.

Find your place on the band. The thin band-plan strip along the bottom of the screen (just above the status bar) colour-codes the CW/Digi/Phone segments around you — pick your IARU region (or leave it on Auto) in the settings drawer → **Band-plan region**. The framed window on the strip is a **slider**: drag it (or tap anywhere on the strip) to move precisely to any spot on the band, and the spectrum, waterfall and VFO all follow along. You can also grab that slider from anywhere along the bottom status bar and drag sideways — a taller target — while a vertical up-swipe there still opens memory channels.

Read your passband. Two grey vertical lines on the spectrum mark your current filter edges. The BW label in the top bar shows the active width; tap it to choose 2.5 / 2.7 / 2.9 / 3.2 kHz in USB or LSB. A coloured tint fills the passband so you can always see exactly what slice of the band you're receiving.

Watch the S-meter. The Signal field in the top bar is a live tick-scale bar (S1 through S9+20), driven by the actual signal under your VFO cursor.

Flat-spectrum mode. Toggle **Flat spectrum** in the settings drawer to switch from absolute dBm to dB-above-local-noise-floor. Noise collapses to a calm baseline; real signals — including weak CW tones in the mud — pop sharply above it. Recommended on noisy bands.

Save your spots. Swipe up from the bottom edge to open the memory channel picker — a 4×8 grid of 32 channels. It doesn't greet you with blank cells: a fresh device arrives with a handful of **example channels already filled in**, and each one is **colour-coded by mode** (CW green, DiGi teal, USB brick-red, LSB purple) so you can read the whole grid at a glance without opening a single label. The very first time you open it, a quick **guided animation** walks you through the gestures — watch a channel slide to a new slot, then drop into the wastebin and vanish — so you learn "drag to move, drag to delete" in about ten seconds, once, and never again.

Then it's yours: **tap** a channel to recall it (the QMX retunes instantly), **tap an empty slot** to store the current frequency/mode, **long-press** to edit, **drag** a channel to reorganise the grid, and **drag onto the wastebin** (bottom-right cell) to delete it.

Monitor from another room — or operate remotely. Once WiFi is configured, open `http://<tab5-ip>` in any browser for a full-featured live panadapter with mouse-driven tuning, band/mode controls, and ADIF log download. The IP is shown in the bottom status bar. See Web UI.

Step 6 — FT8 (if that's your thing)

Swipe in from the **left edge** to switch to the FT8 screen. The Tab5 starts decoding 15-second slots immediately — no PC, no WSJT-X required.

1. Tap the **Preset** button and pick your band's conventional FT8 dial frequency (e.g. 14.074 MHz for 20 m).
2. Watch the decode list fill. CQ stations appear at the top sorted by SNR; exchanges and replies below.
3. To reply to a station: **hold your finger on their row** for ~250 ms. A dim highlight appears after ~80 ms so you can see which row you're targeting before the gate fires. Lift — a confirmation modal shows the exact message before anything is armed.
4. Tap **Auto Pounce** to hand the full QSO to the auto-engine (works through report → RR73 → 73 with patient retry), or **Transmit** for a single manual message.
5. To call CQ: tap **Call CQ**. The engine picks a clear audio slot, fires CQ, automatically answers the first caller, runs the full exchange, logs the QSO, and resumes calling CQ.

Every completed QSO is written to an ADIF log downloadable from the web UI. See FT8 Transmit for the full picture, including ARRL Field Day exchange mode and a no-radio-keyed practice/simulation mode.

Step 7 — Something not working?

Spectrum flat, top bar shows `---`:

- Check the cable between Tab5 and QMX — almost always a charge-only cable.
- Make sure the QMX is powered on and not stuck in its own bootloader.
- Power cycle in order: Tab5 first, then QMX.

For anything else:

1. The diagnostic log is **always on** — nothing to enable.
2. Reproduce the problem (let it run a minute; power-cycle the QMX if the issue is about connection).

3. Grab the log:
   - **Over WiFi:** in the web UI take **Files ▲** → **Diagnostic download** ↓. That one item fetches **both** logs — `qmx-log.txt`, the current session held in RAM, and `qmx-log-saved.txt`, the copy persisted to flash that survives a reboot or a flat battery. If the device restarted, the saved one is the interesting file. (Directly: `/api/log` and `/api/log/saved`.)
   - **microSD:** if a card is inserted, the log is mirrored to `/qmx-panadapter/qmx-log.txt` on the card — continuously with WiFi off, or up to the start-up backup with WiFi on (alongside the always-on flash-persisted copy, which is complete either way).
   - **Over USB (no WiFi needed):** capture the serial console with `tools/capture_serial_log.ps1`.
4. Open an issue and attach the log. It includes Tab5 and QMX firmware versions plus every CAT command exchanged — usually enough to pinpoint the problem immediately.

Getting help

This whole guide is inside the firmware. No WiFi, no microSD card, no download: it works on the first boot, in a field with no signal, and it can never describe a different version than the one you are running. Swipe ← from the right edge to open the settings drawer; the top two buttons are the two ways in.

User Manual — opens where you are. Not a contents page to search: it opens the chapter covering the screen you were on — the panadapter chapter, the FT8 **receive** chapter, or the FT8 **transmit** chapter if a transmission is armed or running (someone mid-transmission is asking a different question). Inside: **Contents** is a two-column list — press and slide, and lifting your finger opens the highlighted chapter; **Back** returns to the previous page (shown only when there is one); **Exit** puts you back where you came from. Edge swipes and top-bar taps are stood down while the manual is open, so a stray touch cannot retune behind it. Holding the **User Manual** button for 3 s resets the reader; the manual itself is part of the firmware and cannot be lost.

Need guidance? — describe the symptom, not the cause. The second drawer button opens a short list headed "*What do you need help with?*", written the way you would say it out loud — "My radio is not showing up", "Nothing appears in the decode list", "It never transmits", "How do I change what my CQ says?", "Where are my contacts logged?" Pick the one that fits and the manual opens at the section that answers it. The list holds questions as well as faults, and it scrolls — keep going past the first few rows.

Rows the Tab5 can see are happening now are highlighted and floated to the top: no CAT link to the QMX, IQ mode never confirmed, an empty decode list while FT8 is running with the radio present, or WiFi switched on but not connected. WiFi switched **off** deliberately — POTA, battery — is not reported as a fault; that row stays as a normal question. **The device ranks, you choose:** a highlighted row is a suggestion, and it will never navigate for you on inference. The rows offered are scoped to the screen you opened the panel from, so FT8 never offers you spectrum symptoms.

Warnings you can tap. A warning you cannot act on is half a warning. The red **"QMX IQ mode not confirmed"** banner across the top of the screen is a button — tap it for the section on that exact fault. The **"Waiting for QMX"** message carries a small **Need help?** button; it says *help* rather than *what is wrong* on purpose, since a radio that is off is often off deliberately.

If a chapter does not answer your question, that is a documentation bug worth reporting — the wording of these rows is written from what operators actually say.

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2. Panadapter

The panadapter is your primary view — a real-time spectrum analyser and waterfall for the QMX.

2.1. Layout

```
type: stack
title: The panadapter screen, top to bottom
row: 60 Top bar | frequency, mode, filter width, S-meter, zoom
row: 200 Spectrum | green trace, amber VFO cursor, tinted passband
row: 32 Frequency axis | absolute MHz labels
row: 370 Waterfall | newest row at the top, SDR gradient
row: 22 Band-plan strip | CW / Digi / Phone segments, with the visible span marked
row: 36 Status bar | battery, UTC clock, WiFi
note: amber | the VFO cursor sits at the dial frequency; in USB the passband tint runs
note: dim | the top-right 200 x 120 of the spectrum is a deadzone, so a tap near the dr
```

Status bar indicators:

Left to right: battery, the SD dot, the firmware version, the UTC clock, and the WiFi indicator with the network name and IP address.

Item	Meaning
Battery % and voltage	Charge level, with the measured pack voltage beside it (e.g. `88% (8.1V)`)
SD (green dot)	A microSD card is mounted and being mirrored **continuously** — the case with WiFi off (see Settings)
SD (yellow dot)	A card is present and your start-up backup is written, but continuous mirroring has stopped — the case with WiFi on. Your log, config and LoTW certificate are safe on the card; later QSOs are written at the next start-up
`UTC(GPS)` / `UTC(NTP)` / `UTC(FT8)` / `UTC(FT4)`	The time source **currently** in charge — GPS-disciplined QMX, WiFi/SNTP, or FT8/FT4 offline sync (RTC / manual show as `UTC(RTC)` / `UTC(MAN)`)
Firmware version	Currently flashed firmware version
WiFi fan icon	Link strength, at the right-hand end with the network name and IP address. The dot alone means a weak link (above 25 %), plus the first bow above 50 %, plus the second above 80 %; all three faint means connected but very weak, or `off`. From v1.3.4 this replaces the old `-NN dBm` figure — the width went to the network name, which is far more often too long to fit

2.2. Touch Controls

Tap to Tune

Tap anywhere on the spectrum or waterfall to jump to that frequency. The tap snaps to a frequency grid so you land on a sensible frequency rather than wherever your fingertip happened to be.

With a mouse the rule is simply the colour of the pointer: **white means clicking tunes, green means you are over something that will act instead** — a callsign, a top-bar control, an edge handle. Press and hold, or click and hold, and drag if you want to place the cursor before committing; a finger and a pointer behave identically.

!!! note "Fixed in v1.8.1" Before v1.8.1 a band across the upper part of the spectrum quietly refused to tune — the touch areas behind the top bar had been made shallower without the tune code being told, so that strip belonged to nobody. It looked as though tuning only worked near the centre frequency, while the waterfall was fine.

> Callsigns from Live Spots are drawn over the spectrum and are the one exception: tapping a **callsign** tunes to that station and sets the mode, rather than tuning to the point you touched.

The snap step follows the **mode**, not the zoom level — a CW signal needs to be found to the hertz, an FT8 dial frequency does not:

Mode	Snap step
CW / CW-R	10 Hz
USB / LSB	250 Hz
DiGi / FT8 / FT4 / RTTY	500 Hz
AM / FM	1 kHz

RIT — receiving off your transmit frequency

Someone answering your call a couple of hundred hertz off your frequency is the ordinary case this is for: you want to hear them without moving your own transmit frequency, which everyone else is listening on.

1. Tap **RIT** at the top right of the spectrum. It turns amber and reads *click a signal* (in the browser) or waits for a tap.
2. Tap the caller on the spectrum or waterfall. The dial does **not** move. A dashed magenta line appears where you are now listening, in the spectrum and down the waterfall, and the filter window moves onto them.
3. It stays armed, so the next caller is another tap.
4. Tap **RIT** again to clear the offset and switch it off.

If you never use RIT, the button can be hidden: **Settings → Radio → Show RIT button** (also in the web settings). It is shown by default. If an offset is actually engaged the button appears regardless of that setting — the radio listening away from your dial is not something to leave unsaid on screen.

The gold line goes on marking the **dial** — which is where you transmit — so seeing both lines at once tells you transmit has not followed your receiver. Tuning anywhere by any means clears RIT, on the assumption that you have moved on.

While RIT is armed, taps use a 10 Hz grid whatever the mode. The normal grid above exists to land the dial on a tidy frequency; RIT is a fine adjustment onto one caller's tone, and SSB's 250 Hz step would leave only a handful of usable offsets. The range is ±500 Hz — tapping further away clamps to the limit and says so.

Available in the browser too, where a click does the same thing. The offset itself is shared, so setting it on either screen shows on both; the *armed* state is per-screen, because arming changes what a click does and a click in a browser is not a finger on the Tab5.

The grid is anchored to absolute frequency, not to where your finger first touched, so the cursor always lands on the same set of points no matter where the drag began.

Pinch to Zoom

Use two fingers to pinch in (zoom out) or pinch out (zoom in). The display centers on the passband width — in USB/LSB modes, the passband center stays on screen even when the VFO is off to the side.

Pan (One-Finger Drag)

Drag horizontally with one finger to scroll the spectrum left/right. Release to tune to the new center frequency.

Band-Plan Strip

Along the bottom of the screen — just above the status bar, below the waterfall — is a thin coloured strip showing where you are within the band at a glance. Each colour zone marks the conventional segment (CW, Digi, Phone) for your selected region (auto / ITU Region 1/2/3, set in settings).

The **visible-span block** inside the strip shows the exact portion of the band the spectrum and waterfall are currently displaying — it narrows as you zoom in and shifts as you pan.

A **passband sub-block** inside the visible-span block mirrors the current filter width at band scale (grey tint), so you can see how your passband sits within the CW/Digi/Phone zones.

Tune directly from the strip:

- **Tap** anywhere on the strip to jump to that frequency
- **Drag** to scrub along the band — the frequency label updates live and the QMX retunes on release

Drag from the bottom bar too. The visible-span block acts as a slider handle that reaches *below* the thin strip: touch on or just under it — anywhere along the bottom status bar — and **drag sideways** to scrub the band, exactly like dragging the strip itself. This gives you a much taller grab target. It coexists with the memory-picker gesture on the same row: a **sideways** drag retunes the band-plan, while a **vertical up-swipe** still opens the memory picker.

The strip updates live as you zoom, pan, or change bands.

Memory Channels

Swipe ↑ from the bottom edge to open the memory picker — a 4×8 grid of 32 channels, each free to hold any frequency/mode (not tied to a band). Each channel stores:

- Frequency
- Mode (USB, LSB, CW, DiGi)
- Label (your own name for the channel)

Each button shows the mode colour (CW = green, DiGi = teal, USB = brick red, LSB = purple) so you can tell mode at a glance without reading the label.

Recall: Tap a filled channel to immediately retune to it.

Tap an empty slot to create a new channel there directly — the frequency keypad opens straight away.

Long-press + drag a filled channel to move it to a different empty slot — no editing needed, the data follows your finger. Release on an empty slot to drop.

Long-press in place to open the editor (change name, frequency, or mode).

Entering a frequency outside the legal amateur band edges is rejected immediately with an error — the pad stays open so you can correct it without starting over.

2.3. Top Bar

Tap any item to open its selector:

Item	Purpose
14.074	Frequency. Tap to open the frequency keypad.
USB	Mode. Tap to open the picker: USB, LSB, CW, DiGi — plus **AM** once the connected QMX reports 1.04 or newer firmware.
2.5 kHz	Bandwidth (SSB only). Tap to choose 2.5, 2.7, 2.9, or 3.2 kHz.
S7	S-meter. Shows received signal strength. Display only — there is nothing to tap.
1.0x	Zoom level. Tap to choose x1, x2, x4, x8, x16 or x24. Pinching sets any value in between.

2.4. Spectrum & Waterfall

The **spectrum** shows signal power in dBm (default range −130 to −30 dBm) as a green curve with a dim fill. Each frame is smoothed per-bin with an exponential moving average — α = 0.4 by default, adjustable in the drawer — which balances a stable picture against a snappy response to CW keying and SSB attack transients.

The **frequency axis** shows absolute MHz labels centred on the QMX VFO, refreshed on every CAT frequency update. At high zoom the labels resolve to kHz or Hz precision.

The **waterfall** runs newest row at the top, in a thermal SDR palette (black → dark blue → teal → green → yellow → red). Four colour maps are available in the drawer: **Thermal, Viridis, Turbo** and **Grayscale**.

The waterfall floor tracks the band automatically. Its black level follows a running median sampled only from bins *inside the passband*, EMA-smoothed, so the background colour follows conditions instead of sitting at a fixed anchor. Bins outside the passband are drawn darker and excluded from that calculation, so they cannot wash out dim in-band signals.

Flat-spectrum mode (drawer toggle, persisted) switches both spectrum and waterfall to a per-bin adaptive display: every bin renders as dB *above its own running noise floor*. Real signals — weak CW tones especially — stand out sharply against a calm baseline, which is why it is the mode to reach for on a noisy band. The dB-range sliders have no effect while it is on, and the axis shows relative dB above floor rather than absolute dBm.

The picture is cleared when it would otherwise mislead. Switching between Panadapter, FT8 and FT4, or changing band, clears the waterfall and resets the spectrum baseline, so stale signals from the old band cannot linger in the new one. The noise floor recalibrates and a new baseline settles in about a second.

Waterfall controls (in settings):

- **Black level** — how far above noise-floor to go black (default 9 dB)
- **Contrast** — dB span of the colour ramp (default 45 dB)
- **Adaptive floor** — blends between a per-bin and a global noise floor. **This has no effect at present** and is not offered in the browser: the per-bin floor is re-seeded many times a second, so the two values it blends are always the same. Left in place because the stored value is still exported with your configuration.
- **FFT window** — Blackman-Harris, Hann, or Nuttall (default Blackman-Harris)

2.5. Frequency Keypad

Tap the **frequency** on the top bar to open the keypad. Enter frequency in MHz format:

- **14.074** for 14.074 MHz
- **1.832** for 1.832 MHz
- **14.074.250** for 14,074,250 Hz — a third group is hertz within the kilohertz

Each group after a `.` is padded to three digits, so `14.07` is 14.070 MHz, not 14.007. Type a number with **no `.` at all** and it is taken as plain hertz.

The digit layout toggles between **10 Key** (calculator order, 7-8-9 on the top row) and **Phone** (1-2-3 on the top row). Pick whichever your fingers already know — the choice persists across reboots.

Resize the keypad: Pinch it or swipe up/down on it to toggle between the normal and a compact layout. Your choice is remembered across reboots. The compact layout reveals more of the spectrum behind it.

Reposition the keypad: Drag it by the **"Enter freq"** title label (the only non-button area at the top of the panel) to move it anywhere on screen, clamped to stay fully visible. The position is remembered so it reopens exactly where you left it.

The background behind the keypad is intentionally semi-transparent (40% dim) so the spectrum and waterfall stay visible while you're entering a frequency.

Cancel and Enter are the only exits — tapping outside the keypad does nothing, so an accidental background touch cannot silently discard what you were typing.

2.6. Memory Channels

32 memory channels (4×8 grid), free to hold any frequency/mode — not tied to a band. Each stores frequency, mode, and name. Swipe up from the bottom edge to open the memory picker.

Each slot shows its **label in large text**, with the mode and frequency dimmed beneath. Mode is also colour-coded — CW (green), DiGi (teal), USB (brick red), LSB (purple) — so the grid is scannable at a glance without reading every label.

To recall a channel: Tap it. The QMX retunes immediately.

To create a new channel (empty slot): Tap the empty slot — the frequency/mode picker opens directly, pre-filled with the current VFO so you can adjust both before naming it. No long-press needed.

To edit an existing channel: Long-press it in place. Change the name, frequency, or mode, then tap **Save**.

To move a channel to a different slot: Long-press and drag it to any empty slot. Release to drop. Only the data moves — the grid positions stay fixed.

To save the current QMX frequency/mode to a slot:

1. Swipe ↑ to open the memory picker
2. Long-press the slot you want to overwrite
3. The current frequency/mode pre-fills the editor — adjust the label if you like, then tap **Save**

To delete a channel — drag it onto the wastebin: the bottom-right cell (channel 32) is a **wastebin**. Long-press any filled channel and drag it onto the wastebin to delete it (it fades out). This is the way to clear a slot you no longer want.

Example channels on first use. A brand-new device ships with a few example channels already filled in (rather than 32 blank cells), so the grid is explorable straight away. These only seed empty slots — they never overwrite anything you've saved.

First-run tour. The very first time you open the memory picker, a brief (~10-second) one-time animation shows that channels can be **dragged** to move them and **dropped on the wastebin** to delete them. It plays once and never again.

Out-of-band frequencies are rejected immediately — if you enter a frequency outside a recognised amateur band, the keypad stays open so you can correct it rather than silently saving a wrong value.

2.7. Band Presets

Tap the **band name** on the top bar to switch between configured bands. The band selector shows:

- **160m, 80m, 60m, 40m, 30m, 20m, 17m, 15m, 12m, 10m, 6m** — standard bands (whichever your radio has configured)
- **11m** — the CB segment (26.9–27.5 MHz), shown on QMX+ radios that expose it
- **Custom** — user-added bands (via settings)

On radios with many configured bands (QMX+), the picker lays the bands out in **two side-by-side columns** so every band is visible without scrolling. Selecting a band jumps to the **nearest** band centre, so closely-spaced bands like 10 m and 11 m are no longer confused for one another.

Switching bands **remembers the last frequency you visited on each band**, so you can flip between 20m and 40m without losing your place.

2.8. Zoom & Pan

The QMX sends a 48 kHz-wide slice of I/Q, so **×1 shows 48 kHz** centred on the dial. Zooming narrows that window with a true zoom-FFT — you get finer resolution, not a stretched picture.

The zoom level is displayed on the top bar (e.g. **2.0x**). Tap it to choose a preset:

Zoom	Visible span
×1	48 kHz — the whole slice
×2	24 kHz
×4	12 kHz
×8	6 kHz — CW pile-up detail
×16	3 kHz
×24	2 kHz — individual CW signals well separated

Pinching sets any value in between; double-tap returns to ×1 and re-centres.

Gesture	Effect
One-finger fast horizontal swipe	Pan ("stroll") the view — retunes to the new centre on release
Pinch (two fingers)	Zoom ×1.0 – ×24.0
Two-finger drag	Pan the zoomed window
Double-tap	Reset zoom and pan to ×1.0, centred
Top-bar **Zoom** → tap	Pick a preset

One-finger pan (stroll). A fast horizontal swipe — more than about 70 px of movement within the first 250 ms of touching down — slides the spectrum and waterfall under your finger in real time, with a live frequency tooltip, and retunes to wherever you release. It works at any zoom level, alongside the two-finger pinch and pan, and it is the quickest way to move along a band without dropping into a deliberate tune-drag.

At zoom levels above ×1 the display **centres on the passband**, not the VFO dial — which matters in USB and LSB, where the passband sits to one side of the carrier. It re-centres automatically when you change mode or filter width, so the active receive area stays on screen, and the passband lines and frequency axis track correctly at every zoom level.

The zoom level is **persisted**, and comes back at full zoom-FFT resolution on the next boot.

2.9. S-Meter

The **Signal** field in the top bar is a tick-scale bar labelled S1, S3, S5, S7, S9, +10, +20, with a moving green bar beneath it.

- **S1 to S9** — standard S-units, −130 to −73 dBm
- **+10 / +20** — above S9, in 10 dB steps

The scale is **S9 = −73 dBm**, 6 dB per S-unit below S9 and 1 dB per unit above it.

The reading is the peak level in a ±64-bin window centred on the **IF-shifted VFO bin** — corrected for the QMX's +12 kHz IF offset, so it measures the signal actually under your cursor rather than the DC/local-oscillator spike sitting at the centre of the spectrum. It stays live during FT8 capture.

It is a readout, not a control: there is nothing to tap, and there is no peak-hold mode.

2.10. Settings Drawer

Swipe ← from the right edge to open the settings drawer, or tap the right grip handle.

It is grouped — **Station, Device, Radio, Network, Display, FT8, Spectrum** — with a **Basic / Expert** toggle at the top. Basic shows what a normal session needs; Expert reveals the tuning and calibration controls, so the list stays short until you need it to be long.

Every control, group by group, is documented once in Settings — deliberately in one place rather than summarised here as well.

Next: Learn about FT8 Receive or Web UI.

Touch-to-tune

Tap anywhere on the spectrum or waterfall to place the cyan tune cursor. Drag and the cursor snaps grid-point to grid-point — you can see exactly which frequency will be tuned before you lift. The snap grid is **mode-aware**:

Tune vs. pan — how your finger is read. A quick horizontal swipe (more than ~70 px within 250 ms) pans the view instead of tuning — see Zoom & Pan. A slower touch-and-hold (250 ms without that much movement) locks into tune mode, after which dragging moves the snap cursor as described below. In short: swipe fast to pan, hold-then-drag to tune.

Mode	Snap step
CW / CW-R	10 Hz
USB / LSB	250 Hz
FT8 / DiGi / RTTY	500 Hz
AM	1 kHz

The grid is anchored to absolute frequency (e.g. …200 / 300 / 400 Hz), not to your touch start point, so the cursor always lands on the same set of grid points regardless of where your finger first touches.

A floating frequency tooltip above the cursor shows the target frequency in real time while dragging. Lift → CAT `FA` command is sent; QMX retunes; spectrum re-centres.

Taps always tune to exactly where you touched (snapped to the mode-aware grid above). The old **Snap to signal** option — which hunted for the strongest bin near your tap — was removed in v0.19.4; predictable tuning won.

Passband indicator. Two grey vertical lines mark your current filter edges. A faint coloured tint fills the passband. The amber VFO marker shows where the QMX is tuned; in CW mode it sits at dial + CW pitch offset so it marks the actual received tone frequency, not the suppressed carrier.

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3. Live spots

Live spots put other stations **on your spectrum**, at the frequency they are actually operating on, so you can see who is where without leaving the radio.

Two sources feed the same display:

Source	What it tells you	Default
POTA (Parks On The Air)	Park activations currently spotted, any mode	**On**
RBN (Reverse Beacon Network)	CW stations the skimmer network is hearing right now	**Off** (opt-in)

Both need WiFi. Nothing here transmits, and nothing here touches the radio until you tap a spot.

3.1. What you see

Spots are drawn **over the spectrum**, the way a FlexRadio or similar SDR shows them — the callsign sits around the middle of the spectrum with a thin vertical line dropping from it down to the frequency axis, so the line points at the frequency the station is on. The line is only 2 px wide, so the trace stays readable underneath.

Colour tells you what a spot means:

Colour	Meaning
Amber	A POTA activation
Bright green	An RBN (CW skimmer) spot
Grey	You have **already worked this station on this band**

Grey is the useful one: it answers "do I need this station?" at a glance. It is band-aware, so the same operator on a different band is *not* greyed out — that is a new band-slot.

Spots fade as they age. A spot is a claim about *now*, and an old one pointing at an empty frequency is worse than no spot at all:

- Under 5 minutes old — full brightness
- Around 15 minutes — about half faded

- 30 minutes — gone

The callsign and its line always fade together.

Off-screen counts. When there are spots on your band that fall outside the window you are looking at, a small count appears in the bottom corner of the spectrum — `< spots (3)` on the low side, `spots (5) >` on the high side. It says "spots" in words rather than a bare `<3`, because a lone number against the edge of the spectrum tells you neither what it counts nor that you can tap it. These are **scoped to the band you are on**, so the arrow never points at something on 40 m while you are on 20 m. Each is coloured like the spot it will take you to.

Crowded bands. Only so many callsigns fit legibly side by side, so on a busy segment the closest-spaced spots lose their name — and a spot with no name is not drawn at all, rather than leaving a line pointing at nothing. Stations you have **not** worked are labelled in preference to ones you have, and fresher spots in preference to older ones.

3.2. Tapping a spot

Tap a callsign and the radio tunes to it *and* switches mode:

Spot mode	Radio goes to
CW	CW
FT8, FT4, other data	DiGi
SSB	USB above 10 MHz, LSB below
Unknown	Frequency only — the mode is left alone

Bandwidth is deliberately **not** forced. The QMX keeps a filter per mode and loads it when the mode changes, so the right width follows by itself.

Tap an off-screen count (`< spots (3)` / `spots (5) >`) and you jump to the nearest spot on that side, which brings it into view. Those counts have a deliberately large touch area — the visible text is small, but the target around it is not.

Tapping the spectrum anywhere else still tunes normally, and pinch-zoom and the one-finger pan are unaffected. Only the callsigns themselves and the two counts are spot targets.

A callsign's target is the text, and nothing above or below it. That matters because a long callsign is a wide label sitting over the trace, and a click inside it goes to *that station* rather than to the frequency under your pointer. If a signal you want is behind a label, aim at the same column higher or lower in the spectrum and it tunes there normally.

In the browser, hovering a callsign shows you a marker at that station's real frequency with its name, so you can see where the click will take you before you commit. On the Tab5 the two gestures are already distinct: tuning to a point needs a brief hold, while a callsign is a quick tap.

3.3. Turning it on and off

Settings drawer (swipe in from the right edge) — **Live spots** section:

Control	Default	What it does
Live spots (POTA)	On	The whole feature. Off leaves the spectrum completely clean
Add RBN (CW skimmers)	Off	Adds RBN as a second source. Needs your callsign set
DX cluster spots	Off	Human-typed spots — the only source of SSB activity. A second connection, so opt-in
SOTA spots (summits)	Off	Summit activations from spothole.app, a volunteer-run server. Opt-in for that reason
Mode filter the spots	**On**	Shows only spots you can work in the mode you are in

Every one is saved and appears in the configuration file you can download and restore from the web UI, under `[settings]`.

About the mode filter, since it is on by default and changes what you see: tapping a spot sets the **mode** as well as the frequency, so an unfiltered lane can drop a CW operator into FT8 without warning — which is exactly what happened to Michael KZ4LY, and why this exists. With it on, changing mode changes which spots are on screen, and the off-screen counts change with them. A spot whose mode is unknown is always shown, so nothing is hidden on a guess. Switch it **off** and every spot is shown, with a two-letter tag on the label (`CW`, `SB`, `FT`) telling you what tapping it will do — deliberately absent when the filter is on, because then every spot is already your mode.

The spots overlay belongs to the **Panadapter** page. It is not drawn on the FT8 screen.

3.4. About RBN

RBN is **off by default, on purpose.** It is a continuous global feed rather than an occasional fetch, and it arrives over a persistent connection on the part of this board that has historically been the most delicate. Off by default means it can never affect anyone who has not asked for it.

If you do switch it on:

- **Your callsign is required.** The feed asks for one when connecting; it is how RBN attributes load, not a password. Set it under **Callsign & Grid**.
- **Only your current band is kept.** RBN reports the whole world, and the display can only show one band at a time, so spots outside the band you are on are discarded as they arrive. Changing band clears the picture and it refills within a few seconds.
- **Duplicates are merged.** The same CQ is typically reported by ten or more skimmers; you see one entry per station, at the best reported signal.
- **RBN spots** are held for up to **10 minutes** after any skimmer last heard them — shorter than the 30 minutes above, because an active caller is re-spotted every couple of minutes and silence means they have stopped.
- It is the **CW/RTTY** skimmer feed. RBN runs a separate FT8/FT4 feed that the panadapter deliberately does not subscribe to — you are already decoding those yourself, far better, from the antenna.

3.5. About the DX cluster — where phone spots come from

RBN cannot show you an SSB station. It is a network of **automated skimmers**, and a skimmer decodes CW, RTTY, FT8 and FT4 — machine-readable modes. There is no such thing as an SSB skimmer, because no machine can recognise a callsign spoken into a microphone. So every phone station on the band is invisible to it.

A **DX cluster** is people typing. That is the whole difference, and it is why this exists as a third source:

- **Phone spots** — SSB activity you cannot see any other way.
- **Park and summit references** — a human writes `POTA ES-2081` in the comment, so a cluster spot can also tag your chase in the log.
- **Everything else people bother to spot** — DX, contest activity, anything worth telling other operators about.

Switch it on with **DX cluster spots (phone)**, next to the other two sources in the settings drawer, or from the web UI. Like RBN it is **off by default**: it is a second continuous connection, and the same caution applies.

Things worth knowing:

- **Your callsign is required**, for the same reason as RBN — the node asks for one to identify the connection.
- **Only your current band is kept**, again like RBN.
- **Mode is worked out for you.** Most cluster spots carry no mode field at all. If the spotter typed one (`LSB`, `CW`, `FT8`) it is used; otherwise the frequency is looked up in your band plan — 14.020 is CW, 14.285 is phone. The same inference you would make reading the line yourself.
- **Skimmer spots relayed onto the cluster are ignored.** Some nodes forward RBN traffic; those entries would duplicate what RBN already gives you, so they are dropped rather than drawn twice.
- Spots are held for up to **15 minutes** — longer than RBN, because a human spot describes something slower-moving than a skimmer's automatic report.

!!! note "Not every node is equally busy" Cluster nodes vary enormously. In testing, one node produced two spots in four minutes while another produced seventy-five in fifteen. A quiet node looks exactly like a broken connection, so give it a few minutes before concluding anything.

3.6. About SOTA — summit activations

SOTA spots come from **spothole.app**, an aggregator run by Ian Renton M0TRT, used with his permission. That is worth knowing because it shapes how the Tab5 treats it:

- **Off by default**, so nobody polls somebody else's hobby server without choosing to.
- **Fetched once every two minutes**, which is plenty for activations that last an hour.
- **When it is unreachable, nothing happens.** The spots already on screen stay until they age out normally — no banner, no error, nothing to dismiss. A volunteer server being down for an afternoon is an ordinary state, not a fault in your radio.

Almost every SOTA spot carries a portable suffix — `EA2GM/P` rather than `EA2GM` — because the operator is on a summit rather than at home. That is why the callsign shown is the one that was spotted, exactly as spotted.

> **SOTA is nearly all CW and SSB.** Of a typical sample, none are in a digital mode. So a summit will rarely appear in your FT8 log; the value of the source is seeing where the activity is on the spectrum, and being able to tap it.

3.7. Refresh and timing

- POTA is fetched about **once a minute** — spot rates on that service change on that sort of timescale, and the fetch briefly pauses the web UI's spectrum stream so the two never compete for the link.
- RBN and the DX cluster arrive continuously and are merged into the display every **10 seconds**.
- Ageing is re-evaluated every second, so spots dim and disappear on their own even when nothing new arrives.

If spots never appear at all, check in this order: WiFi connected (bottom bar), **Live spots** switched on, and — for RBN only — a callsign set.

Other stations drawn straight onto your spectrum at the frequency they are actually using — **POTA** park activations, **RBN** skimmer reports, and **DX cluster** spots, which are where phone activity comes from (no machine recognises a callsign spoken into a microphone, so SSB is invisible to RBN).

Grey means you have already worked them on that band. Press and drag to pick one, and lift to tune it with the right mode. One station is one entry, however many sources report it. Each source switches on separately in **Network → Live spots**.

📖 Full chapter: tab5.lav.dk/guide/spots

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4. Web UI

With the Tab5 on WiFi, open `http://<tab5-ip>` in any modern browser. The IP is shown in the bottom status bar on the Tab5.

The browser panadapter is a full-featured view in its own right — not just a window onto the Tab5. On a larger monitor you get more spectrum history, a bigger waterfall canvas, and mouse controls that are faster than touch for precise tuning. It shows live spectrum at ≈10 fps via WebSocket, full waterfall history (~50 s), the same thermal palette and floor maths, a graphical S-meter, and a top bar with Band / Mode / BW / Zoom controls. The bottom bar shows battery percentage + voltage, firmware version, a live UTC clock, and WiFi SSID + RSSI. To its right: download/upload buttons (ADIF, QRZ, eQSL — see QSO logging), **Files ▲** → **Diagnostic download** ↓ for the diagnostic logs, **Config ↓ / Config ↑** to back up / restore / edit all settings (see Config backup, restore & edit), and **Tab5Shot** which opens a live `/ss.bmp` screenshot in a new tab.

microSD file browser (new in v1.3.0). Files → **SD Files** in the bottom bar opens `http://<tab5-ip>/files` — browse the microSD card from any computer without pulling it: download your logs and config backups, upload files, delete. Card access is coordinated with the WiFi link the same way the automatic backup is, so it's safe to use mid-session.

Whole-band plan strip & adjustable split (new in v0.20.0). Along the bottom of the browser view (above the status bar), a colour-coded CW/Digi/Phone strip spans the entire band with a draggable "visible window" (drag or tap to retune) and a VFO marker, mirroring the Tab5's own strip. Drag the divider between the spectrum and the waterfall to give either more room — the split is remembered in the browser. **Tab5Shot** now captures any open pop-up (band/mode dropdown) too, and the frequency keypad is draggable with a standard 10-key layout.

In FT8/FT4 mode the live stream pauses (new in v0.20.0). The browser stops streaming the spectrum/waterfall and shows a notice plus the log and upload controls instead. While you're operating digital modes the stream would compete with the on-device decoder and the WiFi link, so pausing it keeps FT8 decoding and WiFi noticeably steadier. Switch the Tab5 back to Panadapter mode and the stream resumes automatically.

Live TX status, and Call CQ from the browser. In FT8/FT4 mode that same panel carries a **TX status banner** (v1.3.6) mirroring the Tab5's own label — red while transmitting including the "call 2 of 4" counter, amber armed, green on QSO complete, orange on timeout, plus the persistent "CQ stopped after N calls - no answer"; the browser tab title shows a red dot while transmitting, visible even in a background tab. Under it is a **Call CQ** button (v1.5.0, Dennis WN4FLA), so a CQ run that timed out or reached its call limit can be restarted without walking back to the radio. It **confirms first** — it keys the radio, and a mis-click from another room should not put a carrier on the air — then uses exactly what the Tab5 would: the active CQ preset, the current TX tone (honouring TX Hold) and the TXCQ ANY/EVEN/ODD parity, sharing one code path with the Tab5's own button so the two cannot drift. The request is handed to the display task, so the button greys out for a moment; watch the banner rather than the button. A request arriving while the Tab5 is not in FT8 is discarded, never queued. Replies and pounces are still initiated on the Tab5, where the decode list is.

Click or drag to tune. Click or drag on the spectrum or waterfall — a cyan cursor appears with a live frequency readout and commits on release.

Mouse-wheel to pan or tune. Rolling the mouse wheel over the spectrum or waterfall **pans** the view when zoomed in, letting you survey the band without touching the Tab5. At zoom ×1 the wheel **tunes** with mode-aware snap (CW 10 Hz, SSB 500 Hz, DiGi 100 Hz, AM 1 kHz).

Band / Mode / BW / Zoom dropdown pills in the top bar send commands to the QMX via `/api/cmd` — the same effect as using the Tab5 top bar.

Zoom sync. The browser renders the same zoomed window as the Tab5.

QSO log — download, view, edit. Once you have logged at least one completed FT8 QSO, a **"QSO Logs (N) ▲"** menu appears in the bottom bar of the web UI. **ADIF download** ↓ fetches your `qso.adi` file directly. **View / edit log** (v1.3.5) opens the whole log as a table in the browser — click any column header to sort (click again to reverse; sorting by date groups an activation's QSOs together), delete a single record with the ✕ on its row, or **Delete all** to clear the log (it asks you to type `DELETE`, because there is no undo — download the ADIF first if you want a copy). The menu is only shown when the log contains data.

QRZ / eQSL upload. Two more buttons appear alongside the ADIF link once you have logged QSOs — see QSO logging for the full picture.

Config backup, restore & edit

The **Config ↓ / Config ↑** buttons in the bottom bar let you save every setting to a plain text file, edit it on your PC, restore it, or share parts of it.

- **Config ↓** downloads `qmx-config.txt` — a human-readable INI-style file with everything the Tab5 remembers, grouped into sections:
  - `[settings]` — callsign, grid, WiFi SSID/password, CW pitch, IF trim, IQ balance, flat-spectrum, zoom, colour map, brightness, dB range, EMA, GPS toggle, keypad layout, QRZ key, eQSL user/pass.
  - `[cq]` — the three FT8 CQ-message presets and which is active.
  - `[ft8_filters]` — the CQ-run include/exclude filter terms and toggles.
  - `[memories]` — the 32 memory channels, one per line: `slot = freq_hz, mode, label`.
- **Config ↑** picks an edited file and **merges** it back: only the keys and sections present in the file are changed — everything else is left as-is. Unknown keys are ignored (so a file from a newer firmware still loads). Memory channels apply immediately; other settings take effect after a restart.

Three things you can do with it:

1. **Back up before a clean flash.** Hit **Config ↓** first, do the clean/erase flash, then **Config ↑** to restore everything in seconds — instead of re-typing it all on glass.
2. **Edit in a text editor instead of on the touchscreen.** Faster for callsign, grid, CQ messages, or punching in a batch of memory channels.
3. **Share a band plan.** Copy just the `[memories]` section into a message — another operator pastes it into their file and uploads it to get the same channels (their other settings untouched).

> The downloaded file contains your **WiFi password** and **QRZ/eQSL credentials in clear text** (so it works as a full backup). Keep it private; strip those lines before sharing a file with anyone.

Endpoints

Endpoint	Method	Returns
`/`	GET	Browser panadapter (HTML)
`/api/status`	GET	JSON status (see below)
`/api/cmd`	POST	Send Band/Mode/BW/Zoom commands
`/api/log`	GET	Diagnostic log download (`qmx-log.txt`)
`/api/log/saved`	GET	Flash-persisted diagnostic log from before the last reboot/power-off
`/api/adif`	GET	ADIF QSO log download (`qso.adi`)
`/api/adif/clear`	POST	Wipe ADIF log and worked-call cache; resets the QRZ/eQSL/LoTW upload positions
`/api/adif/delete`	POST	Delete one record: `?idx=<n>&call=<CALL>` — both must match the record or it answers 409
`/api/qrz_key`	POST	Save QRZ Logbook API key (body = raw key text)
`/api/qrz_upload`	POST	Upload pending QSOs to QRZ Logbook; returns `{uploaded, failed, error}`
`/api/eqsl_creds`	POST	Save eQSL username/password (JSON body `{"user","pswd"}`)
`/api/eqsl_upload`	POST	Upload pending QSOs to eQSL (batched); returns `{uploaded, failed, error}`
`/api/config`	GET	Download all settings + memory channels as editable INI (`qmx-config.txt`)
`/api/config`	POST	Upload an INI config; merges into NVS; returns `{applied}`
`/ss.bmp`	GET	1280×720 RGB565 BMP screenshot
`/ws`	WS	Binary spectrum stream (~10 fps)

`/api/status`

```
{
  "battery":     { "level": 100, "mv": 8320, "charging": true },
  "wifi":        { "ssid": "MyNet", "rssi": -44, "ip": "192.168.1.213" },
  "freq_hz":     14074000,
  "mode":        "USB",
  "band":        "20m",
  "passband_hz": 2700,
  "signal_dbm":  -87.4,
  "zoom":        1.0,
  "pan_bins":    0,
  "flat_mode":   false,
  "utc_epoch":   1750000000,
  "qso_count":   12,
  "qrz_key_set": false,
  "eqsl_creds_set": false,
  "tab5_fw":     "v0.18.8",
  "qmx_fw":      "1_03_002QMX"
}
```

Polled at 1 Hz by the landing page; safe to consume from monitoring scripts, home automation, etc. `qmx_fw` is read from the QMX via the `VN;` CAT command at link-up (empty until the radio responds). `signal_dbm` is the peak dBm around the IF-shifted VFO bin (null if DSP has no data yet). `qrz_key_set` / `eqsl_creds_set` tell the web UI whether to prompt for credentials before the next upload.

`/ws` — binary spectrum WebSocket

Single-client endpoint. Each binary frame is 1026 bytes:

Bytes	Meaning
`0`	Frame type: `0x01` = spectrum
`1`	Zoom decimation factor (1 = base 1024-bin spectrum, >1 = zoom-FFT with that decimation)
`2–1025`	1024 unsigned bytes, one per bin, quantised −130 dBm (0) to −30 dBm (255)

Bins are pre-shifted server-side so byte 2 is the leftmost screen pixel. Push rate ≈10 fps.

Screenshots

```
Invoke-WebRequest -Uri "http://192.168.1.213/ss.bmp" -OutFile screenshot.bmp
```

Who is hearing me

(needs WiFi) **Miscellaneous → Who is hearing me** asks PSK Reporter which receivers have copied **your** callsign recently, and lists them by distance with the country, bearing, and the signal report they gave you.

This is the reverse of the reports the panadapter *sends*, and the two are separate settings — you can have either without the other. Switch it on in the browser under **Settings → Spots & reporting**; there is deliberately no Tab5 control, because the answer is a table of distances and bearings that wants a screen you are already sitting in front of.

It answers the one question no amount of local processing can: **am I actually getting out?** The valuable case is the mismatch. Stations you can hear that cannot hear you is a transmit-side problem — antenna, power, a bad connector — and from the receiving side alone that looks exactly like a dead band.

It is read-only, and asks at most once every five minutes because that is the rate PSK Reporter requests. An empty list is the normal answer until you have transmitted: nobody can report hearing a station that has not been on the air.

---

5. FT8 Receive

The panadapter includes **on-device FT8 and FT4 decoders** with real-time spectrum waterfall and a live list of heard stations.

5.1. FT8 & FT4 View

Swipe → from the left edge to toggle to FT8/FT4 view. The same decode list and waterfall work for both modes — switch modes via the **Preset** dropdown in the left pane (top).

FT4 notes:

- Decodes refresh roughly twice as fast as FT8 because slots are 7.5 seconds
- Slot countdown shows 7.5 s instead of 15 s

- All filtering, priority, and auto-reply features work identically in FT4
- Time-sync from decoded signal timing works in both FT4 and FT8 (the bottom-bar clock shows `UTC(FT4)` when synced from an FT4 decode)
- **v0.19.4 made FT4 reliable**: a memory-placement fault previously killed decoding on every other slot (FT8's longer slots mostly hid it), the EVEN/ODD slot markers were on the wrong 15-second grid, and the slot clock could visibly jump between slots — all fixed; if FT4 seemed deaf or erratic on an earlier version, update the firmware

You'll see:

- **Decode list** (left pane) — every station heard, ordered so the ones that matter to you are at the top (see Decode List below)
- **Waterfall** (right pane) — same real-time spectrum as panadapter mode
- **Call CQ button** — start a CQ (requires QMX on the air)
- **Filter button** — include/exclude stations, prioritise by SNR/distance, show only CQ callers

5.2. Decode List

The list shows every decoded FT8 message:

Column	Meaning
SL	Slot parity: **E** (blue) or **O** (amber)
CALL	Their callsign
MESSAGE	The full decoded message text
CTY	Country as a 3-letter code (from the callsign prefix)
SNR	Signal-to-noise estimate, colour-banded by strength
DT	Slot-timing offset in seconds, relative to the band — an on-time station reads ~0.0 (v1.3.1)
HZ	The station's audio tone within the FT8 passband (v1.3.1)
KM / MI	Great-circle distance from your grid
BRG	Bearing from your grid
HRD	Times decoded since last appearance

Own call highlight — your callsign is shown in **inverted colours** (red fill, white text) so you spot replies to you instantly.

How the list is ordered. Top to bottom:

1. **The station you are working** (asked for by Don WB0LQW) — during an exchange there are several transmissions each way, and the partner's messages used to sort down-screen where you had to hunt for them. Now *everything* from them stays at the top: their reply to you, their CQ while you are mid-exchange, and a message they send to a third station — which is exactly when you most want to see it. It covers a hand-typed reply as well as an automatic exchange, and matches on their **callsign**, so a third station merely mentioning them is not promoted. It releases the moment the QSO completes (their closing `73` contains your callsign, so rule 2 keeps it up there anyway).
2. **Messages addressed to you** — anything containing your callsign.
3. **CQ calls.**
4. **Everything else, strongest signal first.**

Stale entries — the list is a live picture of who's on frequency now; stations not heard again within **2 minutes** drop off automatically.

During your own CQ run — stations transmitting on *your* transmit slot (which you can't hear while transmitting) are hidden from the list and return when the run ends; other stations' CQ rows are hidden during a run as before.

Nonstandard callsigns — special-event and compound calls now resolve to the full callsign instead of showing `<...>`, once the station has been heard in full.

5.3. Filtering & Priority

Tap the **Filter** button to open the filter modal:

- **Include callsigns** — only show these calls (space or comma-separated)
- **Include callsigns containing** — show any call with these substrings (e.g., `/P` for portable)
- **Exclude callsigns** — hide these calls
- **Exclude if contains** — hide any call matching these substrings
- **Exclude worked before** — skip calls you've already logged QSOs with
- **Show only CQ callers** — hide replies, show only CQ messages
- **Skip TX1** — when you pounce a station, open with your signal report straight away instead of the grid exchange, for a quicker QSO (falls back to the normal grid exchange if the station has dropped out of the decode list)

- **Allow grey-listing** — off by default. A station that times out two of your pounces in a row is set aside: the robot and Auto-work-pileup skip it, its decode row turns **violet**, and tapping it offers to clear it from the grey-list instead of opening the TX dialog. Handy on a busy band where one station simply never comes back. The list is held in memory only and forgotten at power-off
- **Auto-reply priority** — Strongest SNR, Weakest SNR, Most distant grid
- **Robot mode** — auto-answer CQ (disabled by default; see FT8 Transmit)

All filters persist across sessions. Toggle any filter on/off to enable or disable it without erasing the criteria.

5.4. Decoding Performance

On a reasonably busy band, expect roughly **15–20 distinct messages per slot** in either mode. That is messages, not contacts — one QSO produces several.

Decoding uses **both processor cores**: the decode task works one half of the candidate list on core 1 while a helper takes the other half on core 0, and both run against a waterfall that was built during the capture rather than after it. That is what makes the results appear a second or two into the following slot instead of most of the way through it.

What limits it:

- **LDPC error correction is heavy**, and most candidates the search finds are false alarms that use the full iteration budget before being discarded
- **The spectrum and waterfall** are being drawn from the same audio at the same time
- **USB audio must be serviced continuously** — a gap there costs decodes far more than a slow decoder does

FT4 advantage: because FT4 slots are half as long (7.5 s vs 15 s), you get refresh feedback roughly twice as fast, which feels snappier on a crowded band even though the per-slot decode count is the same.

The panadapter prioritises **receive latency** (decode every 15 s, not accumulate) over volume.

5.5. Real-Time Waterfall

Unlike WSJT-X (which buffers a whole 15-second capture), the Tab5 **builds the waterfall on-the-fly** as audio arrives. You see CW and other traffic in real time, and FT8 stations appear instantly as their symbol blocks are decoded.

The waterfall also shows **non-FT8 activity**:

- CW (carrier lines)
- SSB (broader vertical smears)
- RTTY (FSK sidebands)
- Interference (wideband noise)

5.6. Tap to Reply

Tap any FT8 row in the decode list to **prepare a reply**:

1. A confirmation modal pops up showing your message
2. You can review the target callsign, grid, and SNR
3. Tap **Transmit** to send, or **Cancel** to abort

The reply always follows FT8 protocol (correct parity, proper message sequence) — you can't accidentally send a garbled or out-of-order message.

Pile-up list. If a station answers you while you're busy with another contact, they used to disappear from the list once they stopped transmitting. They're now remembered in a **Pileup** list — the **ADIF-log** button on the FT8 screen turns into a **Pileup** button (a different colour) whenever callers are waiting. Tap it to see who's called you, tap a station to work them, or tap the ✕ to dismiss one. See FT8 Transmit for the full workflow.

5.7. Auto-Reply (Robot Mode)

⚠️ **Experimental** — enabled via a checkbox in the Filter modal.

When robot mode is on, the Tab5 **automatically replies to CQ** without waiting for you to tap. It scans each FT8 slot for CQ callers matching your filters, picks the highest-priority station, and sends a full QSO exchange (TX1 → wait for report → TX2 → wait for RR73 → TX3). Everything is logged to ADIF.

Important: Robot mode **keys the QMX for real** — your signal goes on the air. Never leave it unattended unless you're confident in your filters and your station is in a safe state.

5.8. Search & Window

The decoder uses an **FFT-based search window** to find candidate tone blocks, at whichever tone spacing the active mode uses — **6.25 Hz** for FT8's 8 tones, **20.83 Hz** for FT4's 4.

Frequency stability: The QMX's USB audio clock isn't bit-exact 48 kHz, so decodes slide slowly over time. The panadapter **re-locks to UTC boundaries every 15 seconds** (the slot boundary), preventing long-term drift.

5.9. Time Sync for FT8

FT8 requires **UTC time accurate to ±1 second** for slot synchronisation. The panadapter gets time from:

1. **WiFi + SNTP** (if available) — most accurate, syncs every ~hour
2. **Tab5 RTC** (if set) — accurate to ±1 min, no internet needed (POTA)
3. **QMX GPS** (if QMX has GPS) — re-syncs the RTC every 5 minutes
4. **Manual** — you can set time via the settings drawer

See Time Sync for details.

Next: Learn about FT8 Transmit or go back to Panadapter.

Live FT8 reception on 20 m (v0.15.7). Left pane: MODE / Preset freq / UTC / slot countdown with parity and progress bar / TX parity preference / operator identity / Call CQ / decode summary. Right pane: scrollable decode list.

Swipe in from the **left edge** to switch to the FT8 screen. The Tab5 decodes 15-second FT8 slots directly on the ESP32-P4 — no PC, no WSJT-X.

Prerequisites

- **WiFi configured** — recommended for accurate UTC time. Without it the Tab5 falls back to its supercap RTC or the QMX clock. See Time sync.
- **Callsign and grid** set in the drawer → **Identity** if you want distance/bearing columns or plan to transmit.

The FT8 screen

Left pane: MODE label · **Preset** frequency button (tap to pick a band's conventional FT8 dial frequency) · UTC clock · 15-second slot countdown with current parity (EVEN blue / ODD amber) and a gliding progress bar · **live mini occupancy strip** (the audio window as 50 Hz cells, same data and colours as the tone picker) · **TXCQ ANY / EVEN / ODD** parity-preference button (one button, cycles) · **TX nnnn Hz** tone button · **Call CQ** button · last-slot decode summary and TX status.

Right pane — decode list:

Column	Content
SL	Slot parity: **E** (blue, :00/:30) or **O** (amber, :15/:45)
CALL	Extracted callsign
MESSAGE	Full decoded FT8 message text
CTY	DXCC entity as a 3-letter code (ISO alpha-3 where it exists; ~190 entities)
SNR	FFT-based estimate, colour-banded: green ≥0 / white −5..−1 / orange −15..−6 / grey <−15
DT	Slot-timing offset in seconds, relative to the band — an on-time station reads ~0.0
HZ	The station's audio tone (its offset within the FT8 passband)
KM	Great-circle distance from your grid
BRG	Bearing from your grid
HRD	Times decoded since last appearance

CQ calls always appear at the top sorted strongest-SNR first; all other rows follow by SNR descending.

Row colour scheme (CALL + MESSAGE columns): **white** for ordinary traffic, **green** for plain `CQ` calls, **dim grey** for callsigns already worked **on the current band** (a station you worked on 20 m still shows normally on 40 m — see Reply filter for hiding them entirely), and **inverted red fill + white text** for any message containing your own callsign — your highest-priority rows literally pop off the screen instead of relying on red-on-black text, which field testing found hard to read at a glance.

Live view. Stations not re-decoded within 60 seconds drop off the list automatically, even while the band is quiet — the list is who's on frequency *now*, not a cumulative history. (The "Active: N" counter that used to say so was dropped in v1.3.4; the wrapping TX/QSO status text took its line.)

Performance

On a busy 20 m FT8 slot the decoder regularly yields 25–50 callsigns per slot. Both EVEN and ODD slots decode every cycle via a ping-pong dual-buffer architecture — a TX slot never causes the opposite parity to be skipped. Heap is stable across long sessions (≈39 KB internal RAM free, 25 MB PSRAM free during decoding).

---

6. FT8 Transmit

The panadapter **keys the QMX and transmits full FT8/FT4 QSOs** — reply to CQ, run your own CQ, auto-answer (robot mode), or conduct a full exchange.

6.1. FT8 vs FT4 — Which to Use?

Both modes transmit via the same CAT interface; the difference is **slot length and symbol rate**:

Feature	FT8	FT4
Slot length	15 seconds	7.5 seconds
Symbols transmitted	79	105
Symbol duration	160 ms	48 ms
QSO time (full exchange)	~90 sec (6 slots)	~45 sec (6 slots)
Use case	Weak signal, SOTA/POTA	Busy bands, contests
Preset picker	Panadapter mode → Freq dropdown	FT8 mode → Preset picker

Choose FT4 when:

- The band is crowded (faster exchanges, less QRM)
- You're in a contest (time is critical)
- You want quicker pileup resolution

Choose FT8 when:

- Signal is marginal (longer decode window tolerates more QRN)
- You're portable/POTA (less pressure to transmit fast)
- You prefer the relaxed pace

FT4 has full feature parity with FT8, including time-sync from decoded signals (fixed in v0.19.3 — the time-sync modal shows "FT4" when appropriate and the bottom-bar clock correctly reads `UTC(FT4)` while in FT4 mode).

6.2. Switching Between FT8 and FT4

In FT8 screen (left pane):

- Tap the **Preset** dropdown (shows current frequency/mode)
- Select an FT4 frequency to switch to FT4
- Select an FT8 frequency to switch back

In Panadapter screen:

- Tap the **Band preset dropdown** (top left)
- Some bands have both FT8 and FT4 frequencies listed
- Tap the FT4 variant to switch

Your **settings are sticky** — when you switch back to FT8 or FT4, the panadapter remembers your last frequency, bandwidth, and filter settings for each mode.

The **decode list and pileup are cleared** on the switch (v1.3.3). The two modes use different slot lengths, so stations decoded under the old timing describe a band the new mode cannot hear — and leaving them on screen meant you could tap one and try to work a station that was no longer there.

6.3. Modes of Transmission

6.0.1. Reply to a CQ

In FT8 view, tap a CQ row in the decode list. A confirmation modal appears:

```
type: panel
title: Confirm FT8 Transmission
row: K9ZZ EN52   -07   2138 Hz
row: W1AW FN31   -12   1406 Hz   <- the row you picked
row: SV1ENG JN37 -09   1869 Hz
row: TX1: W1AW OZ1LAV JO45
buttons: Cancel!, Auto Pounce, Transmit+
note: dim | drag up or down the list to move the highlight before you lift your finger
note: amber | Transmit sends this message once; Auto Pounce runs the whole exchange fo
```

- **Transmit** (green) — send this one message on the next correct slot
- **Auto Pounce** (blue) — handle the entire exchange automatically (TX1 → wait for report → TX2 → wait for RR73 → TX3 73)
- **▲ / ▼ Nudge** — move the target to the row above or below, without closing the modal and redoing the selection gesture
- **Cancel** — abort

Transmit is intelligent (v1.3.0) — it builds the correct *next* message from what that station last sent, exactly like a WSJT-X double-click: their CQ → your grid (or your report, with Skip TX1 on), their grid → your report, their report → `R`+your report, their `R`-report → `RR73`, their `RR73`/`73` → `73`. You can run a whole QSO one Transmit tap at a time — and sending the closing `RR73`/`73` **logs the QSO to ADIF** just like an automatic contact. Auto Pounce is offered on any first reply; once you're mid-exchange with a station, its rows offer Transmit only.

A quick tap is enough — you do not need to hold a row before releasing. Hold-and-drag still works for selecting across rows in a busy list.

The reply follows **correct FT8/FT4 parity** — if you're replying to an even slot, you transmit on the odd slot, and vice versa. For FT4 the countdown timer correctly counts down in 7.5-second slot increments, not 15-second FT8 ones.

Skip TX1 (faster pounce). With **Skip TX1** enabled in the Filter modal, pouncing a station opens with your signal report straight away instead of the grid exchange — saving one round trip. If the station has already dropped out of the decode list, it falls back to the normal grid exchange automatically.

If your target is working someone else, the panadapter holds instead of keying up over their exchange — the status shows "working *call* - waiting" with a **TAP TO CANCEL** line. Waiting costs nothing (a hold doesn't count toward the timeout), but if you'd rather move on, tap the status to cancel the pounce and pick a different station; the abandoned exchange stays resumable for a few minutes via the resume prompt.

Working a pile-up

When you call CQ (or work a busy run), more than one station may answer at once — and a caller who replies while you're mid-QSO with someone else used to vanish from the live decode list as soon as they stopped transmitting. They are now kept in a **Pileup** list so you don't lose them:

- Whenever one or more stations are waiting, the **ADIF-log** button on the FT8 screen becomes a **Pileup** button in a distinct colour, and reverts once the list empties.
- Tap **Pileup** to see everyone who has called you and isn't worked yet.
- **Hold the button to open the ADIF log** — the log stays reachable even while the button reads "Pileup" (a one-time hint appears the first time it flips).

- Tap a station in the list to work them (the same confirmation modal as a decode-list tap), or tap the small ✕ to dismiss them without working them. The reply is the correct **next** message for that station, built from whatever they actually last transmitted — usually a signal report for someone who has just called you, but an `R+`report if they came back later with a report of their own (v1.3.2; the same laddering as tapping a decode-list row).
- A station is removed from the list automatically once you start a QSO with them, and a just-completed contact's trailing 73 can't put them back.
- Stations you've worked before still appear in the pileup unless **Exclude worked-before** is checked — the pileup follows the same rule as the auto-answer.

The pile-up tracker never transmits on its own — it only remembers callers; you choose who to work. If you'd rather it *did*, check **Auto-work pileup** in the Filter modal: the strongest waiting caller is pounced automatically when your current QSO completes — or immediately, if you enable it with callers already waiting and nothing else in progress. It carries the same unattended-TX warning as the robot.

6.2. Call CQ

Tap the **Call CQ** button. A modal opens:

```
type: panel
title: CQ Messages (check the active)
row: [x] CQ OZ1LAV JO65
row: [ ] CQ DX OZ1LAV JO65
row: [ ] CQ POTA OZ1LAV JO65
buttons: Cancel!, + call grid, Save+
note: dim | CQ stop and Listen sit top-right and apply on the tap - no Save needed
```

Choose a preset (or edit/save a new one), then tap **Transmit**. The QMX starts calling CQ and listening for replies.

CQ tone selection — the panadapter automatically picks a quiet frequency (6.25 Hz tone spacing, 200–2800 Hz audio range, no other stations nearby). If your chosen tone gets busy during a QSO, the panadapter **auto-relocates to the nearest clear frequency** to avoid stepping on other stations.

CQ auto-stop — by default the panadapter keeps calling until someone answers. If you'd rather call a few times and then pause (a common courtesy on a quiet band), long-press **Call CQ** and tap the **CQ stop** button at the top-right of the preset editor: it cycles through never / 1 / 2 / 3 / 4 / 5 / 10 calls and applies immediately, no Save needed. While calling, the TX status shows the progress ("call 2 of 4"); after the last unanswered call the panadapter listens through one more receive slot (an answer to your final call still starts the QSO normally), then stops and goes idle. Applies to every CQ run, including the automatic resume after a completed or timed-out QSO — each fresh CQ sequence starts the count over.

Listening slot - while you are transmitting you are deaf to your own time window, so the occupancy picture for the window you transmit in is the one that goes stale. The **Listen** button in the preset editor (under **CQ stop**) can spend one slot receiving after every 3, 5 or 10 calls. Off by default, because it changes your on-air cadence, and like CQ stop it applies on the tap with no Save needed.

6.3. Auto-Reply (Robot Mode)

⚠️ **Requires explicit checkbox in the Filter modal** to enable.

When robot mode is on and you're idle, the panadapter scans every FT8 decode for CQ callers matching your filters, picks the highest-priority match (Strongest SNR, Weakest SNR, or Most distant grid — your choice), and starts a full QSO:

1. **TX1** — send your call, grid
2. **RX** — wait for their report
3. **TX2** — send their report + your call back
4. **RX** — wait for RR73 or 73
5. **TX3** — send 73 or RR73 (exchange complete)
6. **Log** — ADIF entry created, move to next CQ

A busy pick is abandoned, not waited for (Roy KI0ER's reasoning adopted verbatim): when *you* pounce on a station that turns out to be mid-QSO with someone else, the Tab5 politely holds — a deliberate pounce means you want *that* station. But the robot picked its target off a list, so when its pick engages a third station the robot **moves on to a different CQ caller** instead of idling through someone else's QSO. No grey-list strike: busy is not unresponsive.

Each QSO takes **~90 seconds** (6 FT8 slots × 15 s/slot). If a station doesn't respond in 4 slots, the panadapter times out and resumes scanning CQ.

Important: Robot mode **transmits on the air unattended**. Never enable it unless:

- Your station is physically attended
- Your filters are tested and correct
- Your antenna/SWR is known good
- Your QMX power is set appropriately

A permanent on-screen warning appears whenever robot mode is available.

6.4. FT8 Simulation Mode (Practice QSOs)

⚠️ **For practice only** — no real stations involved, radio never keys up. Since v1.3.0 the simulator needs **no QMX connected at all** — no radio, no antenna, zero RF.

Enable **FT8 Simulation Mode** in the settings drawer to practice everything: manual step-by-step Transmit, Auto Pounce, CQ-runs, pileups, and Field Day exchanges. Useful for:

- Learning the full FT8 exchange sequence
- Testing your setup without risking interference
- Verifying ADIF logging works correctly
- Practicing the pileup tools on a genuine pileup

How it works:

- **Six phantom stations** (three US, three DX) call CQ on their own tones — every message is real FT8, synthesized to actual GFSK audio and decoded through the same receive pipeline real RF uses
- Tap a CQ to pounce (auto or fully manual — they answer either), or **Call CQ** yourself and **four phantoms answer at once**, building a real pileup
- Phantoms are **patient like real operators**: each repeats its message every cycle, up to four times, before giving up and going back to CQing; one you're working stops CQing; one you've worked stops answering your CQs (toggle sim off/on to reset)
- Their replies match **what you actually transmitted** — grid gets a report, a report gets a roger, `RR73` gets a courtesy `73`
- The **Fast pounce** toggle is honoured, so you can watch what it changes: decodes surface just before the slot boundary with it on, just after with it off
- Swiping to the Panadapter and back clears the phantom rows and pileup for a fresh session, and **turning simulation mode off also clears them** (v1.3.3) — phantom stations no longer linger in a list that is supposed to be real, where they were still tappable

!!! note "Fixed in v1.3.3" With **Fast pounce** turned *off* — which is the default — the phantoms never answered at all, so every practice pounce simply timed out. If you tried the simulator before v1.3.3 and concluded the phantom stations were ignoring you, that was this, not you.

Visual indicator: When simulation mode is active, a **10 px red border pulses around the entire screen** (breathing red frame). This is your visual reminder that you're in practice mode — if the red frame is gone, simulation is off and real stations are in play.

The interlock is in the firmware, not just the UI. While simulation is on, every CAT command that would key the radio is skipped and logged instead — so a QMX that happens to be connected is never keyed, regardless of what the screen is showing.

Important:

- The QMX is **never keyed** in simulation mode, no matter what
- QSOs are logged as real ADIF entries — deliberately, so the logging/upload paths get exercised too. When you're done, the ADIF viewer shows a **"Del N test"** button (only while practice contacts exist — they're recognized by their missing frequency): two taps deletes them all
- All other features (panadapter, web UI, settings) work normally

6.5. ARRL Field Day Mode

During ARRL Field Day, tick **Field Day mode** in the Filter modal and fill in the two fields beside it:

- **Class** — your number of transmitters plus the category letter, written together: `16A`, `5B`, `1D`. It is not a power rating.
- **Section** — your ARRL/RAC section, e.g. `EMA`, `WCF`, `NNJ`.

This switches the FT8 exchange from grid-and-signal-report to Field Day's class+section format, using the standard FT8 message type WSJT-X uses for FD (`WA9XYZ KA1ABC R 16A EMA`). Both pounce and CQ-run follow the convention automatically: the opening message is unchanged — FT8's CQ format has no room for class and section — and the *report-equivalent* step carries them instead, with the receiving side echoing back `R`-prefixed.

While the mode is on, **Call CQ** tags your CQ as `CQ FD <call> <grid>`, using the same modifier mechanism as `CQ POTA` or `CQ DX`, so other Field Day stations know to expect this exchange. It **replaces** any other modifier for as long as the mode is enabled — FT8 allows exactly one. The CQ preset editor reflects that: the three presets are dimmed and locked while FD mode is on, since their own modifier cannot apply, and a live preview line shows exactly what will go out.

Completed Field Day contacts log the standard ADIF contest fields — `CONTEST_ID=ARRL-FD`, your section and theirs — alongside the usual call, frequency and time, so they import cleanly into contest-logging software.

6.6. Fox/Hound (DXpedition) Mode

Big DXpeditions run FT8 in **Fox/Hound** mode, where one station (the *Fox*) works five callers at a time from below 1000 Hz while everybody chasing it (the *Hounds*) calls from above 1000 Hz. It is not the ordinary FT8 exchange: a hound that answers the Fox has to **move onto the Fox's frequency** to do it, because the Fox listens only to its own narrow slice.

Set **Fox/Hound (DXpedition)** in the Filter window to one of three positions:

- **Off** — normal FT8. Nothing changes.
- **Guided** — the Tab5 tells you when it can see a Fox, and you tap it to call. Every transmission stays your decision.
- **Automatic** — the Tab5 calls a Fox it recognises and runs the whole exchange.

Whichever you choose, the awkward part is handled for you: your first call goes out **above 1000 Hz** on a clear slot, and the moment the Fox answers you with a report, your reply is **moved down onto the Fox's own frequency** — which is the step that makes the contact possible at all. The Fox's `RR73` ends it, and the Tab5 then returns to its calling tone ready for the next one.

The Tab5 does not send `73` to a Fox. The Fox's frequency is the scarcest thing on the band during a DXpedition and a courtesy `73` there is just clutter. The contact is complete, and logged, on the Fox's `RR73`.

How a Fox is recognised — and where it trusts you instead. For anything it does *by itself*, the Tab5 is strict: not frequency alone (plenty of ordinary stations work below 1000 Hz) but a station down there visibly working a **queue** — several different callsigns inside a few minutes, a pattern nothing else on the band produces. Until it has seen that, automatic mode will not call anything a Fox.

A tap from you needs no such evidence. While Fox/Hound is enabled, tapping any station in the Fox region runs the Fox/Hound exchange, because enabling the mode is your declaration that you are chasing a DXpedition — the same way WSJT-X's Hound tick works. The machine is deliberately more cautious than you are.

The catch is leaving the mode on after the DXpedition has finished: a tap at an ordinary station low in the passband would then QSY onto their frequency and skip your `73`. So the transmit-confirmation window **tells you before you commit** — it adds a line reading "*as HOUND — will QSY onto 500 Hz, no 73 sent*" whenever Fox/Hound rules are about to apply. If you see that on a station you did not think was a DXpedition, set Fox/Hound back to **Off**.

Three of the Tab5's usual courtesies stand down while a Fox contact runs, and all three would otherwise work against you:

- it **keeps calling** while the Fox is busy with other stations — a Fox is always busy, and waiting for a free frequency means never calling at all;
- it **does not re-send** a closing message, because the Fox never asks for one;
- it **does not grey-list** the Fox for ignoring you, which in a pileup of hundreds is entirely normal.

Automatic mode will not call a Fox already in your log on that band, so it works one contact and then leaves the DXpedition alone.

!!! note "Hound only — the Tab5 cannot be a Fox" A Fox transmits up to five signals simultaneously. The Tab5 keys the QMX one tone at a time over CAT, so a multi-signal transmission is not something this radio can be asked for. This is a hardware limit, not a missing feature.

6.7. Message Status

The FT8 left pane shows a **status label** indicating what's happening:

Status	Meaning
ACTIVE (red)	Transmitting right now
ARMED (amber)	Ready to transmit next slot
QSO Complete (green)	Exchange finished, logged to ADIF
QSO Timeout (orange)	No response after 4 slots, exchange aborted
(dim white)	Idle, no active QSO

Tap the status label to **abort** the current QSO (only works if ARMED or ACTIVE; once COMPLETE, it's logged).

Resume after timeout — if a QSO times out because the partner faded, and they come back within ~5 minutes calling you, the exchange **resumes where it left off** automatically (or tap their row to resume manually) instead of restarting from scratch.

If they never heard your final (v1.3.4) — a partner who does not decode your closing `73`/`RR73` keeps sending you their report, waiting for it. The Tab5 now notices: if the station just worked comes back with a report rather than `RR73`/`73`/`RRR`, the final is sent again, up to three times within four minutes, *before* anything else can start a new contact. The QSO is not logged a second time. Taking over by hand no longer produces a duplicate entry either — the same callsign on the same band inside ten minutes is recognised as the same contact.

6.8. Power & SWR Readout

After each transmit burst, the status bar briefly shows:

```
Last TX: 5.2W SWRx1.25 [N=79]
```

- **5.2W** — average output power measured via QMX `PC;` CAT command
- **SWRx1.25** — SWR measured via QMX `SW;` CAT command (only valid during TX)
- **[N=79]** — number of symbols transmitted
  - FT8: always 79 symbols (~12.7 s at 160 ms/symbol)
  - FT4: always 105 symbols (~5.0 s at 48 ms/symbol)

These readings are **informational only** — the panadapter does not enforce limits. Monitor your antenna system independently.

6.9. CQ Presets

You can save up to 3 custom CQ messages:

1. Tap **Call CQ**
2. Tap **Edit Presets** (or long-press the active preset)
3. Type your message: callsign, modifier (optional), grid
4. Tap **Save**

Presets persist across power cycles. Common modifiers:

- **DX** — calling DX only
- **POTA** — Parks on the Air activation

- **SOTA** — Summits on the Air activation
- *(blank)* — standard CQ

The preset editor's top-right **CQ stop** button sets the CQ auto-stop limit (see Call CQ above) — it cycles never / 1–5 / 10 and applies on tap, independent of Save/Cancel.

6.10. Frequency & Tone Control

Your transmit tone is chosen for you by default — the nearest clear 50 Hz slot, scanned against the stations currently decoded. You can see it, move it, and from v1.3.4 pin it.

The TX tone button is in the FT8 left pane, to the right of the `TXCQ` parity button, and always shows a number (1500 Hz until you change it). Tap it to open the picker.

The mini occupancy strip sits directly under the slot countdown in the same pane: the same 50 Hz grid and the same colours as the picker's full-size strip, so where the band is busy — and where you are in it — is answerable at a glance without opening anything.

Both time windows, always (asked for by Roy KI0ER). The strip is **two rows — EVEN above, ODD below**, marked `E` and `O` in the same blue/orange the slot countdown uses. Two stations only collide if they transmit in the *same* window, and only you know which window you are about to pounce into — so both pictures are on screen **before** you choose, not after a transmission has fixed your window. Your white marker sits on your own window's row once one is locked (both rows until then), and your partner's pink sits in *their* window. The picker's full-size strip is split the same way, and its verdict says where your pick stands: "*Clear in EVEN — busy in ODD*".

In the picker:

Control	What it does
Occupancy strip	The whole 200-2800 Hz window as 52 slots. **Green** free, **red** occupied, **white** you, **pink** your QSO partner
Touch and drag	Drag along the strip to pick a slot. The bar turns grey and follows your finger, the readout tracks it live, and it commits when you lift off
-50 / +50	Nudge one slot at a time
Find clear slot	Jumps to the nearest free slot, scanning outward from where you are
TX Hold	Pins the tone: every CQ and every reply goes out on it. See below
Apply nnnn Hz	Commits the tone and the hold setting

The readout states in words whether your chosen slot is clear or occupied, and lists the nearest free slots as numbers.

It applies **between** bursts. With a burst on the air, Apply refuses and says to try again after it. Slot parity is untouched, so moving your tone mid-QSO does not disturb the exchange — your partner tracks the slot, not the frequency. This is the same freedom WSJT-X gives you, and it is the way out from under a station that has landed on top of you.

TX Hold is WSJT-X's "Hold Tx Freq". With it **off** — the default — each transmission takes the nearest clear slot, and a CQ that gets clashed relocates itself on the next cycle. With it **on**, the tone you picked is the tone used for everything, a clash is reported but never acted on, and nothing moves you off the slot you chose. The line under the checkbox says which of the two you are getting. Both the tone and the hold setting survive a power cycle.

!!! note "The strip shows decoded stations, not raw spectrum" A station too weak to decode will not appear as occupied, and an all-grey strip means nothing has been heard yet rather than "the band is clear".

!!! note "Occupancy is per time window" Two stations only collide if they transmit in the *same* slot, so a tone busy in the odd window may be completely free for you in the even one. The strip accounts for this whenever your own transmit window is known — that is, once a reply or a CQ is armed, since a reply inherits the opposite of your partner's slot and a CQ carries your `TXCQ EVEN`/`ODD` choice. With `TXCQ ANY` and nothing armed there is nothing to work from, so both windows are shown combined. When the window *is* known, the free-slot line names it.

Waiting for a busy station

On a crowded band several stations answer the same CQ and the caller works one of them. If that is not you, the Tab5 no longer keeps calling: while their last decoded message is addressed to somebody else, **nothing is transmitted**, and the status line shows `working <call> - waiting`. As soon as they send `73` or `RR73`, or call CQ again, it picks up where it left off.

You are not committed to the wait (Roy KI0ER's report). The status line carries a **TAP TO CANCEL** line while the hold is on; tapping it drops the pounce so you are free to work somebody else. The abandoned exchange stays resumable for a few minutes, in case the station frees up and you want back in.

This also protects the grey-list. Giving up on a station used to count against it, so a popular station could end up permanently skipped by the robot and Auto-work-pileup for no reason other than being busy. A wait is not a failed attempt. The wait is capped at about six minutes so a station that vanishes mid-exchange still times out normally.

To tune to a different frequency while CQ is running:

1. Tap the **Freq** label on the top bar
2. Enter a new frequency
3. The CQ continues on the new frequency next slot

6.11. ADIF Logging

Every completed QSO is **automatically logged to ADIF** on the Tab5's internal storage:

- **Callsign**, grid, report (SNR), time (UTC)
- **Frequency**, mode (`FT8`)
- **Duration** (QSO start to finish)
- **Operator** (your callsign)

Download the log via the web UI (**QSO Logs** menu → **ADIF download** ↓) or Settings → ADIF Log.

6.12. Upload to QRZ, eQSL & LoTW

Via the web UI:

1. Open the **QSO Logs** menu in the bottom bar and click **QRZ upload** ↑, **eQSL upload** ↑, or **LoTW setup** (reads **LoTW** ↑ once configured)
2. Enter your API key (QRZ), username/password (eQSL), or run the guided certificate setup (LoTW) when first prompted — credentials are saved for future sessions
3. Click again to upload

See Web UI → LoTW Upload for the LoTW certificate setup details.

Logs are batched — each upload session records which QSOs have been sent, so re-running skips already-submitted entries.

Uploads work **while FT8 or FT4 is actively running** — the panadapter briefly steps the FFT and SD-archive activity aside for the duration of the HTTPS transfer, then resumes automatically. You typically miss one FT8 slot and won't notice. A progress result is shown once the upload completes.

6.13. Troubleshooting

"QMX not responding to TX command"

- Check USB cable (must be data cable, not charge-only)
- Verify QMX is in the correct mode (USB/LSB/CW/DiGi, not AM)
- Try a manual `TX;` command via the web UI's CAT tab

"⚠ FREQ BUSY warning, but no other station is there"

- The clash detector reserves a ±50 Hz guard band around each decoded station, so it flags a neighbour that is close rather than exactly on you

- It counts only stations **in your own transmit window** (it used to count both windows, so it could warn about a station that could never collide with you; Roy KI0ER caught it)
- The warning names the frequency. Tap **TX nnnn Hz** and either **Find clear slot** or drag to a green slot on the occupancy strip
- Remember the strip only knows about stations it has *decoded* — a very weak one will not show

"QSO timeout — no response"

- The called station didn't hear you (too weak, bad timing, etc.)
- Try a different station, or switch to manual mode and check levels first
- See Troubleshooting for more

Next: Set up Time Sync for accurate FT8 timing, or explore the Web UI.

The Tab5 transmits FT8 via the QMX's `TA<freq>;` CAT command — no PC audio path, no WSJT-X. The QMX does all DDS synthesis and envelope shaping; the Tab5 sends the 79 tone-frequency commands at 160 ms cadence, bracketed by `TX;` / `TA0;` / `RX;`.

Replying to a station

1. In the decode list, **hold your finger on a row** for ~250 ms. A dim highlight appears after ~80 ms so you can see which row you're targeting. List scroll locks once the gate fires; drag up or down to land on the right row.
2. **Lift** — a confirmation modal shows the exact message that will go on air, the audio frequency, and the target slot parity.
3. Tap **Auto Pounce** to hand the full QSO to the auto-engine, or **Transmit** for a single manual message.
4. Tap **Cancel** (in the modal, or the armed indicator in the left pane) to disarm without transmitting.

Transmit is intelligent (v1.3.0): it builds the correct *next* message for that station from what they last sent — the same semantics as a WSJT-X double-click. Their CQ → your grid (or your report directly, if **Skip TX1** is on); their grid → your report; their report → `R`+your report; their `R`-report → `RR73`; their `RR73`/`73` → `73`. The report value is always your live measurement of their signal. You can walk an entire QSO step by step with nothing but Transmit taps — and when you send the closing `RR73`/`73`, the QSO **logs to ADIF** exactly like an auto-engine contact. Auto Pounce is offered on any first reply; mid-QSO rows get Transmit only.

Slot parity is set automatically — if you heard them on an EVEN slot, your reply goes on ODD so they're listening when you transmit.

The auto-engine works the full exchange: TX1 (grid) → wait for their report → TX2 (R+report) → wait for RR73/73 → TX3 (73) → DONE. At every step it re-sends the current message for up to 4 consecutive slots if the other station doesn't respond. If no reply comes after 4 slots, the QSO times out (orange status, tap to clear).

Skip TX1 (faster pounce). With **Skip TX1** enabled in the Filter editor, a pounce opens with your signal report immediately instead of the grid exchange — saving one round trip. If the station has already aged out of the decode list, it falls back to the normal grid TX1 automatically.

Working a pile-up

When you call CQ or work a run, more than one station may answer at once — and a caller who replies while you're mid-QSO with someone else used to vanish from the live decode list once they stopped transmitting. They're now held in a **Pileup** list so you don't lose them:

- Whenever callers are waiting, the **ADIF-log** button on the FT8 screen becomes a **Pileup** button in a distinct colour, reverting once the list empties.
- Tap **Pileup** to see everyone who has called you and isn't worked yet. Tap a station to work them (the same confirmation modal as a decode-list tap), or tap the ✕ to dismiss one.
- **Hold the button to open the ADIF log** — the log is always reachable this way, even while the button reads "Pileup" (v1.3.0; a one-time hint appears the first time the button flips).
- A station is removed from the list automatically once you start a QSO with them, and a just-completed contact's trailing 73 can't put them back.
- Working a caller from the pileup sends the correct *next* message for that station, built from whatever they actually last transmitted — the same laddering as tapping a decode-list row (v1.3.2). Previously it always opened with a signal report, which could never produce the `R`+report a station needs when they come back minutes later with a report of their own.
- Worked-before stations appear in the pileup unless **Exclude worked-before** is checked — the pileup follows the same rule as the auto-answer, so dupes you're willing to work stay visible.

The tracker never transmits on its own — it only remembers callers; you choose who to work. If you'd rather it *did* transmit, check **Auto-work pileup** in the Filter editor: when your current QSO completes (or immediately, if you check it with callers already waiting and nothing else going on), the strongest waiting caller is pounced automatically, draining the pile one contact at a time. It carries the same unattended-TX warning as the robot.

Calling CQ — CQ-run mode

Tap **Call CQ**. No confirmation modal. The engine:

1. Scans the current decode list for occupied 50 Hz audio bins.
2. Picks the nearest unoccupied bin to 1500 Hz (standard FT8 sub-band centre).
3. Arms the CQ on the matching slot parity and waits for the boundary.

From there it's fully automatic:

- The opposite-parity slot is decoded while CQ runs so replies are never missed.
- The moment a station replies with your callsign, CQing stops and the exchange starts automatically — best-SNR caller chosen if multiple stations answer simultaneously.
- After the exchange completes (RR73 → 73 → logged), CQ resumes on the same frequency.
- A mid-exchange timeout also resumes CQ rather than dropping the frequency.
- While CQ-run is active, other stations' `CQ` rows are hidden so replies to you stand out.

Tap **Cancel** in the TX status bar at any time to stop.

Reply filter

Tap **Filter** in the left pane to open the filter editor. Controls which stations CQ-run auto-answers and which rows appear in the decode list.

- **Include 1 / Include 2** — if either field has text, only messages containing one of its terms are eligible. Space- or comma-separated (e.g. `POTA SOTA` or `JA, VK`), matched against the *whole* decoded message text so POTA/SOTA tags, grid squares, country prefixes, `/P` suffixes etc. all work.
- **Exclude 1 / Exclude 2** — messages containing any of these terms are skipped even if they'd otherwise match.
- **Exclude plain CQ callers** — hides bare `CQ ...` rows so only replies and exchanges remain visible.

- **Exclude worked-before** — hides callsigns already worked **on the current band** (per-band, from your ADIF log) from the list and from CQ-run auto-replies; the same station on a different band is treated as not worked.
- **Skip TX1** — when you pounce a station, open with your signal report instead of the grid exchange for a quicker QSO (falls back to the grid exchange if the station has aged out of the decode list).
- **Allow grey-listing** — off by default. When enabled, a station that times out two of your pounces in a row is set aside: the robot and Auto-work-pileup skip it, its decode row turns violet, and tapping it offers to clear it rather than opening the TX dialog. Useful on a busy band where one station simply never comes back. The list lives in RAM only and is forgotten at power-off.

Tap **Save** to persist (NVS) and apply immediately.

Auto-answer CQ (robot)

⚠ **Transmits unattended — never leave it running unsupervised.** When enabled, the robot picks a CQ caller every slot (filtered the same way as above, plus a worked-before skip) and runs the full exchange itself — no per-QSO confirmation, no tap required. Pick a **Priority**: Strongest signal, Weakest signal, or Most distant grid. Same TX1→report→RR73→73→ADIF-log flow as a manually-tapped reply; you just aren't the one tapping. You remain responsible for everything it transmits under your callsign, same as any other unattended digital-mode software.

TX status indicator

The left pane shows a persistent status line below the slot countdown:

Colour	Meaning
Red	`TRANSMITTING: <message>` — burst in progress, with your **audio tone on its own line**; tap to abort
Amber	`TX armed: <message>` then `nnnn Hz → EVEN/ODD, ~Ns` — waiting for slot; tap to cancel
Red-orange ⚠ FREQ BUSY	Your tone is occupied (±50 Hz guard), and the warning names the frequency so you can act on it. During CQ-run this self-heals — the next cycle hops to the nearest clear tone automatically. Mid-exchange you can now move it yourself: see **TX audio frequency** below
Amber, `working <call> - waiting` + `TAP TO CANCEL`	The station you are calling is mid-exchange with somebody else. Nothing is transmitted until they sign off or call CQ again — or tap to cancel the pounce and work someone else (v1.3.5). See **Waiting for a busy station** below
Green	QSO complete
Orange	QSO timed out — tap to clear
Dim white	FT8 engine status passthrough (capturing / decoding / symbol count / …)

TX audio frequency

Your transmit tone is chosen automatically — the nearest clear 50 Hz slot to 1500 Hz, scanned against the stations currently decoded. From v1.3.3 you can both **see** it and **change** it.

The **TX nnnn Hz** button in the FT8 left pane always shows it — to the right of the `TXCQ` parity button — and tapping it opens the picker. A **mini occupancy strip** under the slot countdown shows the same picture at a glance. (Before v1.3.4 this was a chip that appeared only while a CQ or QSO was running, and the tone was also repeated on the TX status line.)

Control	What it does
Occupancy strip	The whole 200-2800 Hz window as 52 slots. **Green** free, **red** occupied, **white** you, **pink** your QSO partner
Touch and drag	Pick a slot by dragging along the strip. The bar goes grey and follows your finger, the readout tracks it live, and it commits when you lift off
-50 / +50	Nudge one slot at a time
Find clear slot	Scans outward from where you are and jumps to the nearest free slot
Apply nnnn Hz	Sends it to the radio

The readout also says in words whether the slot you have picked is clear or occupied, and the free slots nearest you are listed as numbers underneath.

It applies **between** bursts. If a burst is on the air, Apply refuses and tells you to try again after it — the same rule WSJT-X operators are used to. Slot parity is untouched, so moving your tone mid-QSO does not disturb the exchange; your partner tracks the slot, not the frequency.

> **The strip shows decoded stations, not raw spectrum.** A station too weak to decode will not show as occupied, and an all-grey strip means nothing has been heard yet rather than "the band is clear".

QSO override buttons

During an active auto-QSO exchange (any state between first reply and final 73), three buttons appear in the left pane:

Button	What it does
Re-send / <field> (amber)	Re-arms the current outgoing message immediately. The label shows what will go out — e.g. **Re-send / JO45** when waiting for their report, **Re-send / -07** when waiting for RR73
RR73 (blue)	Skips straight to RR73, bypassing the report step
73 (green)	Fires the 73 sign-off and closes the QSO immediately

These are deliberate operator nudges for busy-band edge cases where the auto-engine falls a slot behind or you can see the exchange is further along than the state machine knows. The auto-engine handles the normal case; these exist for when a quick nudge is faster than waiting through the 4-slot retry cycle.

TX power / SWR readout

After each burst the Tab5 queries `PC;`/`SW;` for instantaneous forward power and SWR and shows the result briefly in the status line: `Last TX: X.XW SWRx.xx [Ns]`. If SWR > 4.0 (indicating the QMX SWR-protection latch tripped), the firmware automatically cycles `TX;` / 150 ms / `RX;` to clear the latch so the radio is ready for the next TX slot without operator intervention.

CQ message presets

Long-press the **Call CQ** button to open the preset editor. Three message slots with radio buttons — check the active one. A `+ <call> <grid>` quick-insert appends your identity. Standard CQ constructions (`CQ DX OZ1LAV JO65FR`, `CQ POTA …`) and ≤13-char free text both encode via the general ft8_lib encoder. Presets persist to NVS. The Call CQ button label shows the selected message; a short tap transmits it.

CQ auto-stop (v1.3.5, Don WB0LQW's "I usually send CQ 2-4 times and then pause"). The editor's top-right **CQ stop** button cycles never / 1 / 2 / 3 / 4 / 5 / 10 calls and applies the moment you tap it — no Save needed. While calling, the TX status shows the counter live ("call 2 of 4"). After the last unanswered call the Tab5 listens through one more receive slot — an answer to your final call still starts the QSO normally — then stops and goes idle. The limit applies to every CQ run, including the automatic resume after a completed or timed-out contact; each fresh sequence starts the count over. Persists across power cycles and travels in the config backup.

QSO logging (ADIF)

Every completed FT8 QSO — pounce or CQ-run — is automatically written to an ADIF v3.1.4 file at `/spiffs/qso.adi` on the Tab5's internal flash. The log survives reboots and re-flashes.

Download. The web UI bottom bar shows the **QSO Logs** menu once the first QSO is logged. **ADIF download** ↓ fetches `qso.adi` for import into any logging software (WSJT-X, Log4OM, DXKeeper, ...) or for POTA.app, which has no upload API (see below); **View / edit log** opens it as a sortable table with per-record delete and Delete-all — see Web UI.

View and edit on the Tab5. The **ADIF Log** button on the FT8 screen (long-press always works, even while it reads "Pileup") opens the on-device viewer: a Today/All filter with a POTA activation counter (the title turns green at 10 QSOs today), long-press a row to delete that single record, and — new in v1.3.5 — **Delete all**, next to Close: the first tap arms it and shows the live count ("ALL 34?"), a second tap within five seconds erases the log, waiting disarms it. Clearing the log (from either the Tab5 or the browser) also resets the QRZ/eQSL/LoTW upload positions, so QSOs logged afterwards upload normally. The clear-first workflow is the practical POTA tool: start the activation with an empty log and the ADIF at the end is exactly what you submit.

Upload to QRZ Logbook / eQSL — directly from the Tab5. Two more buttons appear next to the ADIF download link once you've logged a QSO: **"QRZ ↑"** and **"eQSL ↑"**. Tap either the first time and it prompts for credentials (QRZ: API key from *Logbook Data → API Key* on qrz.com; eQSL: your username and password — eQSL has no API-key scheme) and saves them on the Tab5, not just in the browser. Every tap after that uploads everything logged since the last successful upload — no need to re-enter credentials, and no risk of duplicate uploads, since the Tab5 tracks how far it's gotten into the log. If a record is rejected (bad credentials, quota, etc.) the run stops and reports the reason rather than skipping past it; fix the cause and tap again to resume from where it left off. eQSL accepts a whole batch of QSOs per request; QRZ's API takes one at a time, so a big log takes one HTTP round trip per QSO.

Upload to ARRL LoTW — also directly from the Tab5. The Tab5 signs your QSOs on-device with your own LoTW callsign certificate, so no TQSL and no PC are involved. The bottom bar shows **"LoTW setup"** until it is configured, then **"LoTW ↑"**.

Setup is a two-page guided flow. Page one points you at ARRL's own "Save the Callsign Certificate" instructions — you export a `.p12` file from TQSL, which most people did once, years ago. Page two takes that file, its password, and your DXCC entity number:

Field	Notes
.p12 file + password	Parsed **in your browser**, not on the Tab5 — the password never reaches the device. Only the certificate and key are sent
DXCC entity number	Required. The same one your TQSL station location uses (Denmark = 221, USA = 291)
CQ zone / ITU zone	Optional
US state / county	**US stations only** (v1.3.3). Without them your QSOs earn no Worked All States and no county credit — for you or for the stations you work. The county is the name on its own (`Arlington`), not `VA,Arlington`

Every tap of **LoTW ↑** afterwards signs and uploads everything logged since the last successful upload. Ctrl-click **LoTW ↑** to re-run setup — a certificate is typically valid three years. Re-importing the *same* certificate no longer re-uploads your whole log; only an actually-changed certificate resets the upload position, since a new key means everything has to be re-signed.

> Uploads are queued at ARRL's end. If a QSO does not appear immediately, check the LoTW queue status before assuming a problem — the queue has run hours behind at busy times. LoTW rejects a malformed file at upload time, so anything that reaches the queue was signed correctly.

POTA.app has no upload API at all (browser login or email only), so use the ADIF download above for that.

Fields in each record:

ADIF field	Content
CALL	Their callsign
FREQ	Dial frequency in MHz (3 d.p., e.g. `14.074`)
BAND	Amateur band (e.g. `20M`)
MODE	`FT8`
SUBMODE	`FT8` (required by LoTW TQSL and eQSL for digital mode credit)
RST_SENT	Our SNR estimate (e.g. `-07`). **Omitted entirely if no report was exchanged** — writing a placeholder `599` into an FT8 log, as versions before v1.3.4 did, fabricates a measurement that then gets uploaded to QRZ/eQSL/LoTW
RST_RCVD	Signal report received
QSO_DATE	UTC date `YYYYMMDD`
TIME_ON	UTC time `HHMMSS`
MY_CALL	Your callsign (from Identity in the drawer)
MY_GRIDSQUARE	Your grid
GRIDSQUARE	Their grid (from the decoded FT8 message)
CONTEST_ID, STX_STRING, SRX_STRING, ARRL_SECT, MY_ARRL_SECT	Field Day mode only: contest ID `ARRL-FD`, your/their literal `<class> <section>` exchange text, and their/your section alone

Clear. `GET /api/adif/clear` from the web UI wipes the file and resets the worked-call cache.

> Set your callsign and grid in the settings drawer → **Identity** before your first QSO so they appear correctly in logged records.

7. Time sync

FT8 requires accurate UTC time — within ±1 second of the real thing. The panadapter syncs time from multiple sources in priority order.

7.1. Time Sources (Priority Order)

1. **GPS-disciplined QMX** (auto-detected) — phase-locked to the GPS second, ~10 ms, works offline
2. **WiFi + SNTP** (if available) — ~10 ms, re-syncs every ~1 hour
3. **Tab5 RTC** (if set) — persists across power cycles (±1 min accuracy)
4. **FT8/FT4-derived** — offline fallback only (ignored while GPS/SNTP is up)
5. **Manual set** — enter time manually via the settings drawer

GPS and SNTP are the two accurate sources and are used whenever present; if you're offline with neither, the panadapter falls back to the RTC (or FT8-derived / manual).

7.2. Offline (POTA / Portable)

If you're operating **without WiFi** (POTA, portable, SOTA):

1. **Set the Tab5 RTC before you leave home** (settings → Time)
2. The RTC is powered by a **supercap battery** and holds time for **30–40 hours** without power
3. When you turn the Tab5 on in the field, it reads the RTC immediately
4. Optional: if your QMX has GPS, the panadapter syncs the RTC from QMX GPS every 5 minutes

No internet needed — FT8 timing works offline.

7.3. WiFi + SNTP

When WiFi is active:

1. The panadapter connects to an SNTP server (default pool.ntp.org)
2. Gets the current UTC time
3. Sets the system clock and writes it to the Tab5 RTC
4. Checks periodically (~hourly) and re-syncs if time has drifted

SNTP sync is **automatic** — you don't need to do anything. The bottom bar shows the current time (updates every second).

7.4. QMX GPS Time Sync (auto-detected)

If your QMX has **internal GPS** (QMX+ models often do), there is **nothing to enable** — the Tab5 detects it automatically:

1. At connect, it catches the QMX's `TM;` seconds *flip* (the true GPS second boundary) and compares it against SNTP.
2. If they agree tightly, the QMX is flagged GPS-disciplined and the clock is **phase-locked to that second edge** (~10 ms, drift-free), not just set to the whole second. Re-locks every 5 minutes.
3. If they don't agree (a non-GPS QMX with a free-running RTC), it's used only as a low-priority offline fallback.

The verdict is remembered across reboots, so starting up away from home keeps GPS timing without having to re-detect it. A GPS-disciplined QMX shows **UTC(GPS)** in the bottom bar and is a top-tier source (as good as SNTP); a non-GPS QMX shows **UTC(QMX)** and only helps when offline.

!!! note "The label needs the radio to still be there" The remembered verdict says *this radio is GPS-disciplined*, not *GPS is keeping time right now*. With the QMX unplugged the Tab5 has no GPS of its own, so the label falls back to whatever is actually in charge — usually `UTC(NTP)`. Reported by Don N2VGU, whose Tab5 read `UTC(GPS)` with the radio disconnected; the time was correct throughout, only the label was wrong.

7.5. Setting the time by hand

There is no "Set Manual Time" control in the settings drawer. Earlier versions of this guide said there was, and there never has been on recent firmware — Don WB0LQW went looking for it during a POTA activation and could not find it. Apologies. The clock is set from the FT8 screen:

1. Switch to **FT8**
2. Tap **Filter** (left pane)
3. Tap **Sync Time** (bottom right of the modal)
4. A panel appears with three boxes: **[HH] : [MM] : [SS]**
5. Tap **HH** or **MM** to type them on the numpad (0–23, 0–59)
6. **Hold the SS box and let go exactly on the minute** — the seconds are set to 00 the instant you release
7. Tap **Apply**

Step 6 is the one that matters off-grid. FT8 needs the clock right to about a second, and until v1.8.1 the seconds could not be set at all — hours and minutes were editable and the seconds were not, which left no way to get inside a second with no WiFi and no GPS.

The clock is set **when you release**, not when you tap Apply. The release is the measurement, so an Apply a few seconds later would be a few seconds late. Use a watch with a second hand, or listen for the gap between FT8 transmissions. While you hold the box it turns amber and tells you what letting go will do.

If you have typed hours and minutes first, those are used. If you have not, the Tab5 takes the nearest minute to its own clock — so this corrects the seconds without disturbing an hour and minute that are already right.

The time is written to the RTC as well as the system clock, so it survives a power-off.

7.6. FT8-Derived Sync (Offline Fallback)

The panadapter can also **estimate the UTC offset** from decoded FT8/FT4 signal timing and nudge the clock:

1. Decode several messages (requires on-air activity)
2. Measure their timing relative to the slot boundary
3. Estimate the error and apply a damped correction

This runs only as an offline fallback. While SNTP or a GPS-disciplined QMX is available they are authoritative and FT8-derived sync is **ignored** — deliberately. The FT8 slot offset the device measures is dominated by ~½ second of one-way *receive audio latency* (QMX SDR + USB buffering), which is **not** a clock error; letting it pull the clock would actually drag your transmit timing late. So it only touches the clock when you're off-grid with no SNTP/GPS.

Most useful when: WiFi is unavailable, no GPS QMX, and your RTC has drifted.

Fine-Tune Time with "Sync Time" (FT8 Only)

In **FT8 mode**, the Filter modal includes a **"Sync Time" button** that opens an interactive time-setting panel:

1. Tap the **Filter** button (FT8 screen, left pane)
2. Tap the **Sync Time** button (bottom right of the modal)

3. A panel appears with three fields: **[HH] : [MM] : [SS]**
    - **HH** / **MM** (hours/minutes) — tap to edit via numpad (0–23, 0–59)
    - **SS** (seconds) — auto-syncs from FT8 signals (blue frame = actively syncing, grey = locked). **Hold it and release on the minute** to set the seconds to 00 yourself, which is what you want when there are no FT8 signals to sync from
4. Tap **Apply** to write the time to the RTC and system clock

The **SS field** updates automatically from decoded FT8 messages — each decode gives a sub-second correction estimate. Tap **SS** to toggle between auto-syncing (blue) and locked (grey). While locked, seconds still count at the captured offset, so you don't lose precision after locking.

Use case: You're operating portable without WiFi, your RTC is ~5 seconds off, and there's on-air FT8 activity. Set HH/MM manually, let SS auto-sync to the decoded signal timing for a few seconds, lock it, and you're done — FT8 timing is now precise.

FT8-derived sync shows **SS (sub-second)** in the bottom bar when active.

7.7. Bottom Bar Time Display

The center of the bottom bar shows the current UTC time and which source is active:

Indicator	Meaning	Updates
UTC(GPS)	GPS-disciplined QMX, auto-detected, phase-locked to the GPS second (~10 ms)	Every 5 min
UTC(NTP)	WiFi + SNTP	~1 hour
UTC(FT4) / **UTC(FT8)**	FT4/FT8-derived offline sync	Continuous (when decoding, offline only)
UTC(RTC)	Tab5 supercap RTC	At boot
UTC(MAN)	Manual time set	On-demand
UTC(QMX)	Non-GPS QMX RTC fallback (offline only)	Every 5 min
UTC	Fallback (no sync source)	—

The suffix tells you at a glance which source is **currently in charge** (the active authority, not merely the last one that wrote the clock). A GPS-disciplined QMX (`UTC(GPS)`) and WiFi/SNTP (`UTC(NTP)`) are both accurate — if you see either, you're set. If you're off-grid you'll see `UTC(FT8)`/`UTC(FT4)` (on-air digital activity), `UTC(QMX)` (non-GPS QMX clock), or `UTC(RTC)` (the supercap RTC you set before leaving home).

7.8. Slot Timing

FT8 operates on **15-second slot boundaries** aligned to UTC. The panadapter:

1. Reads the current system time
2. Waits until the next 15-second boundary (hh:mm:00, hh:mm:15, hh:mm:30, hh:mm:45)
3. Starts a 15-second capture window
4. Decodes the received FT8
5. Transmits (if armed) at the start of the next boundary

This alignment is **automatic** — you don't configure slots. But **time accuracy is critical**:

- ±500 ms error → can miss decodes or transmit off-slot
- ±1 s error → very few decodes, transmit often off-time
- ±2 s error or worse → FT8 doesn't work

7.9. Time Sources Summary

Source	Accuracy	Updates	Works Offline?
QMX GPS	~10 ms (auto-detected, tick phase-lock)	every 5 min	Yes (if QMX has GPS)
SNTP	±10 ms	~1 hour	No
RTC	±1 min	at boot	Yes (30–40 h)
Manual	±1 sec	on-demand	Yes (until power cycle)
FT8-derived	offline fallback only	continuous	Yes

Next: Configure Settings or troubleshoot Time Sync Problems.

The Tab5 needs accurate UTC for FT8 slot timing. Sources in priority order (highest first):

Source	When applied
SNTP (WiFi)	Always authoritative whenever WiFi is connected and has synced at least once — sets system clock, Tab5 RTC, and NVS
Tab5 RTC	RX8130CE supercap-backed (~30–40 h retention) — applied at boot before QMX or WiFi are available
QMX `TM;`	Offline fallback only — applied when WiFi is down or has never synced (field/POTA with no WiFi)
FT8 signal timing	Sub-second correction from the decoded signal population's average timing, auto-applied continuously every slot while FT8 is decoding (damped, so a noisy single slot can't yank the clock) — deliberately tracks who you're actually trying to work, not absolute GPS/NTP truth. A manual one-shot version is also available via the **Sync Time** modal (see below)
Manual	Full HH:MM override in the **Sync Time** modal — for rare POTA sessions with neither WiFi nor QMX clock

Each accepted sync writes through to the RX8130CE so the clock persists across power-off. The Tab5 also polls `TM;` every 5 minutes in the background to catch QMX GPS lock events.

For POTA / no-WiFi use: uncheck **WiFi initiated** in the settings drawer. Time comes from the QMX on USB connect (or the supercap RTC if the Tab5 was recently synced). FT8 slot timing will be accurate from either source.

FT8 time calibration modal

On the FT8 screen, tap **Filter** → **Sync Time** to open the time calibration modal. It shows three large boxes: **[HH] : [MM] : [SS]**.

- **HH / MM** — pre-filled from the current UTC clock. Tap either box to edit it with a numpad. Use this for rare POTA situations where neither WiFi nor the QMX clock is available.

- **SS** — auto-syncs continuously from decoded FT8 signals. Each slot, every successfully decoded station contributes a timing sample; a robust outlier-rejecting average (median ± a tolerance window) across all of them sets the correction, so one bad decode can't throw off the reading. The box border flashes bright blue for ~30 ms each time a fresh measurement lands — a visual heartbeat that sync is active — and the hint below reads **"Flash: FT8 synced..."**. A blue frame means it is actively tracking; tap SS to lock it.
- **Apply** — if only SS was synced (HH and MM untouched), `time_sync_apply_correction_ms()` nudges the system clock by the measured sub-second offset and writes through to the RTC. If HH or MM was edited, `time_sync_set_manual()` sets the full time. The bottom bar shows **UTC(FT8)** after an FT8-derived correction, or **UTC(manual)** after a full manual set.

---

8. Settings

Swipe in from the right edge to open the settings drawer. All settings are saved automatically.

The top two buttons are not settings but the two ways into help: **User Manual**, which opens this guide at the chapter for the screen you came from, and **Need guidance?**, which lists symptoms and questions in plain words. See Getting Help.

8.1. Basic and Expert

The drawer is grouped under headings — **Station**, **Device**, **Radio**, **Network**, **Display**, **FT8** and **Spectrum** — and the toggle beside the **Settings** title chooses how much of it you see. It always says where you are and what a tap gives you: tapping **BASIC (tap for Expert)** reveals the Spectrum and Device groups, which hold the calibration and tuning controls you set once and rarely touch again.

Nothing is lost in Basic; it is only hidden. The choice is not remembered between sessions, because it is a way of looking at the drawer rather than a preference.

8.2. Radio

These reach the QMX itself over the CAT link.

QMX volume — the radio's own AF gain, in decibels, the same number the QMX shows on its LCD when you click its volume knob. Read back from the radio when you open the drawer, so it agrees with the rig even if you last changed it there.

QMX RF gain — the per-band RF gain from the radio's Band Configuration, 0–99 dB (the QMX default is 54). Because it is **per band**, the Tab5 reads it from the radio rather than remembering a number that would belong to whichever band you were on last; it shows "reading…" until the radio answers. It writes when you let go of the slider, not while you drag, because this is stored configuration rather than a session setting. Changing it moves the noise floor, so the flat-spectrum reference is re-learned.

CW transmit offset — transmit a little away from the station you are listening to, so a QRP call is not buried in the pile of everyone zero-beating the DX (suggested by Roy KI0ER). The centre of the slider is off, and the sign chooses whether you transmit above or below.

Keep it small: around 100 Hz or less is typical, and the slider stops at ±300. The point is to land inside the other station's filter, not outside it — most CW operators run a 500 Hz filter or narrower, 200 Hz is not unusual, and the passband is not equally usable right to its edge: the tone gets muddy and a QRP signal has least energy exactly where the filter is already attenuating (Michael KZ4LY). Roy KI0ER uses +60 Hz on his own rig. Earlier versions of this page suggested 400–600 Hz, which is outside many operators' filters altogether.

The QMX has no XIT, so this is done with **split**: you receive on VFO A and transmit on VFO B, which the Tab5 holds at your receive frequency plus the offset. Set it once and it follows you — a tap on the panadapter, a spot, a memory recall, a band change, the web page, and the radio's own tuning knob. It applies in **CW and CW-R only**, and leaving CW clears it. If you are running your own split, the Tab5 leaves it alone.

Let me use the QMX menus — hands the radio back. The QMX's own menu, and its Band Configuration terminal application, talk over the same USB serial port the Tab5 polls several times a second, so the two fight over it. Tap this before using the radio's own menus: the Tab5 stops sending anything, the spectrum freezes, and a blue bar across the top says so and gives the radio back when you tap it. Resuming re-checks IQ mode, which a trip through the radio's menu can switch off.

While the radio is released the Tab5 will not transmit — an armed FT8 burst and Antenna Tune are both refused rather than keying a radio you are holding.

8.3. Operator Info

Callsign — Your amateur radio callsign (required for FT8/FT4 logging).

Grid Square — Your Maidenhead grid square (e.g., JO45; required for FT8/FT4 exchanges).

8.4. WiFi

Tap **WiFi setup** in the settings drawer to open the WiFi window.

WiFi On/Off — the WiFi icon button in the WiFi window (shown with a diagonal red slash when off). Toggling it takes effect immediately — off disconnects and stays off; on reconnects right away. Turning WiFi off is useful for field operation (no WiFi overhead, extended battery life).

SSID — Your WiFi network name. Tap **Scan** to list nearby networks and pick one (as of v1.3.6 this also works when your stored network is out of range — hotels, POTA sites — where it previously always reported "No networks found").

Password — Your WiFi password (pre-filled with your saved password; tap the eye icon to show/hide it). Tap **Save** to store it and connect.

Once connected, the settings show your **IP address** — use this to access the web UI from a browser.

8.5. Time Sync

SNTP Server — NTP pool (usually `pool.ntp.org`). Change only if you have a local NTP server.

Setting the time by hand — there is no control for this in the drawer, despite what earlier versions of this guide said. The clock is set from the FT8 screen: **Filter → Sync Time**. See Setting the time by hand, which also covers setting the seconds — the part that matters with no WiFi and no GPS.

FT8-Derived Sync — Estimates the UTC offset from decoded FT8 signal timing. **Offline fallback only**: it is automatically ignored while SNTP or GPS is available (those are authoritative), and only nudges the clock when you're off-grid with no better source. See Time Sync.

QMX GPS — Detected **automatically**, no setting to toggle. If your QMX (typically a QMX+) is GPS-disciplined, the Tab5 recognises it at connect by comparing the QMX's own second-tick against SNTP, and then phase-locks to the GPS second boundary (~10 ms) as an offline time source. On a non-GPS QMX nothing happens. The bottom-bar clock shows **UTC(GPS)** when a GPS-disciplined QMX is the active source.

8.6. Display

Flip 180° — Invert the display for upside-down mounting or cable routing.

QMX volume — The radio's own AF gain, **in decibels: the same number the QMX shows on its own LCD**, not a percentage. Available in both Panadapter and FT8 modes. It reads the radio back each time you open the drawer, so if you turn the volume with the rig's own knob the slider follows instead of disagreeing with it. This exists mainly for QMX+ builds with no control panel, where there is no volume knob at all (Randy N4OPI's request). Nothing is sent to the radio until you move the slider, so switching on can never change your volume unexpectedly.

The slider runs 0–50 dB rather than the QMX's full 0–199 dB protocol range, so the useful part of the travel is not crammed into a couple of centimetres. 50 dB is Randy N4OPI's figure from a real radio with the antenna connected — an earlier "not loud enough" reading turned out to have been taken with the antenna off, where the missing loudness was band noise, not gain. If you turn the radio past 50 dB with its own knob the slider knob sits at the end of its travel, but the number still shows the radio's true dB — it has to agree with the LCD.

Two things worth knowing, both from Randy N4OPI's side-by-side check against a real QMX (the numbers do read identically): the QMX only shows the figure on its own LCD when you give its knob a click, and the Tab5 slider reads the radio when the drawer *opens* — so if you turn the rig's knob while the drawer is on screen, the slider catches up the next time you open it rather than tracking live.

Display sleep — Dropdown (Off / 1 / 2 / 5 / 10 / 30 min). After the chosen idle time the backlight turns off; FT8, the radio link, and the web UI keep running. Tap the screen to wake — the wake tap is swallowed, so it can't tune or press anything. A **two-finger double-tap** blanks the display immediately.

Brightness — Screen brightness (0–100%).

Spectrum Mode —

- **Normal** — absolute dBm scale
- **Flat** — relative to per-bin noise floor (signals pop above baseline)

8.7. Battery Care

Charge Limit — Optionally stop charging once the battery reaches a set percentage (default **80%**), to reduce long-term wear on the pack. Enable it and choose the target in the settings drawer; charging restarts automatically if the level later falls well below the target (5% hysteresis). Leave it off to always charge to 100%.

Accurate charge reading while charging — the displayed battery percentage (and the charge-limit trigger) now compensate for the voltage rise that occurs while charging current is flowing. Previously this made the reading jump around while plugged in, and made the charge limit either stick just short of the target or oscillate; both are fixed.

8.8. Waterfall Controls

Black Level (dB) — How far above noise floor to show as black (default 9 dB). Lower = more colour detail.

Contrast (dB) — Span of the colour ramp (default 45 dB). Lower = more contrast, higher = more gradation.

Adaptive Floor (%) — Blend between per-bin noise floor (100%) and global mean floor (0%; default 100%).

FFT Window —

- **Blackman-Harris** (default) — best frequency resolution
- **Hann** — smoother peaks
- **Nuttall** — sharpest edges

8.9. Panadapter & Zoom

Distance in Miles — Show FT8 distances in miles instead of km (off by default).

Band-plan region — Sets which region's band plan drives the coloured CW/Digi/Phone strip along the bottom of the screen. **Auto** derives it from your grid square; you can also force Region 1/2/3.

Band Presets — Add or remove custom bands. Standard bands (160–10 m) are always available.

8.10. Bluetooth mouse

A Bluetooth mouse is the one pointer that works **while the QMX is plugged in**. A USB mouse cannot: the radio occupies the Tab5's only USB host, and sharing it through a hub does not work on this hardware — both devices end up disabled. A Bluetooth mouse never touches that port, so the radio keeps its connection.

For anyone operating with cold or unsteady hands, that is the point: every menu, button and drawer control becomes a click instead of a precise tap on glass.

!!! warning "The mouse must be Bluetooth 4.0 or later" Only **Bluetooth Low Energy** mice work. The Tab5 gets its Bluetooth from a co-processor that has no Bluetooth Classic radio in it at all, so an older Classic mouse cannot be made to work by any firmware change.

```
Most mice sold since about 2014 are Low Energy. Look for **Bluetooth 4.0** or
later on the box. A dual-mode mouse works on its Low Energy channel.

**How to tell which problem you have.** A Classic mouse never appears to the
Tab5 at all — the Bluetooth symbol keeps looking and never turns blue, because
the Tab5 listens only for mice that announce themselves the Low Energy way. A
mouse that *does* connect but moves the pointer oddly is a different matter,
and that one is worth reporting.
```

Setting it up

1. Tick **Bluetooth mouse** → **Enable** in the settings drawer.
2. **Restart the Tab5.** Bluetooth can only start once the radio link to the wireless co-processor is up, so the switch takes effect on the next boot — the toast says so at the time.
3. Put the mouse into pairing mode and wait a few seconds. It connects on its own.

You only do this once. The pairing is stored, survives a reboot and a firmware update, and the mouse reconnects by itself from then on.

The Bluetooth symbol in the bottom bar sits just left of the WiFi indicator and has three states:

Symbol	Meaning
Dim grey	Bluetooth is switched off
Pale	On and looking for a mouse
Blue	A mouse is connected

What works: moving the pointer, left click, and the scroll wheel — the wheel scrolls whatever is under the pointer, so the decode list, the settings drawer and the manual all scroll.

!!! note "The pointer disappears when the mouse sleeps" A Bluetooth mouse switches itself off after about half a minute of not being moved, to save its battery. The pointer vanishes with it and comes back the instant you touch the mouse — reconnecting takes under a third of a second. This is the mouse looking after its own battery, not a fault.

8.11. Live Spots

Draws other stations on the spectrum at the frequency they are operating on. Full details, including what the colours mean, in Live Spots.

Live spots (POTA) — On by default. Fetches Parks On The Air activations about once a minute and draws them over the spectrum. Needs WiFi. Switching it off leaves the spectrum completely clean.

RBN spots (CW skimmers) — **Off** by default, and deliberately so: RBN is a continuous global feed over a persistent connection, unlike POTA's occasional fetch. Adds stations the skimmer network is hearing right now, filtered to the band you are on. **Requires your callsign** to be set (the feed asks for one on connect — it identifies you, it is not a password).

DX cluster spots (phone) — **Off** by default, same reasoning: a second persistent connection. This is the source that carries **SSB**. Skimmers are machines, and no machine recognises a callsign spoken into a microphone, so phone activity is structurally invisible to RBN however long you leave it on. A cluster is people typing, so it carries voice contacts, and park and summit references written into the comment. Mode is worked out from the spotter's comment, or from your band plan when the comment does not say. Relayed skimmer spots are dropped, so the cluster cannot double what RBN is already showing.

Whatever you switch on, **one station is one entry**: the same callsign within 2 kHz is merged, an activation spot outranks a plain sighting, and a dropped duplicate folds back in as corroboration — drawn brighter, meaning a receiver copied them just now rather than someone having typed it an hour ago.

8.12. FT8 Settings

FT8 On/Off — Globally enable/disable FT8 mode.

Field Day Mode — ARRL Field Day mode (on/off, class, section).

Simulation Mode — Practice QSOs with six phantom stations — no QMX needed at all, radio never keyed (red breathing border on screen). See FT8 Transmit.

Fast pounce (early decode) — On by default. Decodes surface ~1.8 s *before* the slot boundary (WSJT-X style), so a fresh CQ can be answered in the very next slot and mid-QSO replies land on the beat. Trade-off: the capture window closes early, so a station transmitting late in the slot can occasionally be missed. ⚠️ *Not yet A/B-verified on a live band — if your decodes-per-slot drop with it on, turn it off and please report your numbers.*

Distance in miles — Show the decode list's distance column in miles instead of kilometres.

Report to PSK Reporter — **On by default.** Uploads the stations you decode to PSK Reporter, the same as WSJT-X, so you appear on the map as a monitoring station and other operators can see where they were heard.

What is sent, over the internet only and **never on the air**: your callsign and grid square, and for each station you decode their callsign, their grid (if their message contained one), the frequency, the signal report and the mode. Reports are batched and sent at most once every five minutes.

It does nothing at all until both your **callsign and grid** are set, and it is disabled outright in **Simulation Mode**, so practice contacts can never reach the public map. Uncheck this box to switch it off entirely.

> The first report goes out 5–5½ minutes after switching on, so nothing appears immediately. To check it is working, look for **QMX Panadapter** under "Software in use" at pskreporter.info/cgi-bin/pskstats.pl.

FT8 Filters — Include/exclude stations, set auto-reply priority, enable robot mode, grey-listing (see FT8 Receive for details).

Keyboard — M5Stack Tab5 snap-on keyboard support (if connected).

8.13. Audio & DSP

IQ Balance — Adaptive I/Q phase correction (usually on). Suppresses mirror-image signals.

8.14. Diagnostic Logging

The diagnostic log is **always on** — there is nothing to enable. All firmware log output is captured to a 5 MB memory ring, with a rolling copy persisted to internal flash (survives a reboot or power-off) and, if a microSD card is inserted, mirrored to the card as well (see microSD Auto-Archive for when the card copy is written — the flash copy is always complete). Download via:

- Web UI: open the **Files** menu in the bottom bar and click **Diagnostic download ↓** (downloads both the live session log and the flash-persisted copy from before the last reboot)
- microSD card: `/qmx-panadapter/qmx-log.txt`
- USB serial: `tools/capture_serial_log.ps1`

Useful for troubleshooting rare issues.

8.15. microSD Auto-Archive — Station Backup

Insert a microSD card (FAT32 or exFAT, any size — a plain 32 GB FAT32 card is ideal) **before switching the Tab5 on** and it automatically mirrors your whole station to `/qmx-panadapter/` on the card. It's a **grab-and-go backup**: pull the card into a PC (or another Tab5) to back up or move your setup — no computer needed in the field.

File	Contents
`qso.adi`	ADIF QSO log — after each new entry with WiFi off, otherwise at the next start-up
`qmx-config.txt`	All settings + memory channels, as INI text (restore via **Config** upload)
`lotw_cert.b64`, `lotw_key.b64`	Your LoTW signing certificate + private key, so a restored device can sign for LoTW
`qmx-log.txt` (+ `.1`)	Diagnostic log, rolling (rotated at 5 MB)
`README.txt`	A plain-text description of every file, written on each mount

Insert the card before switching the Tab5 on. A card pushed in later is not picked up until the next start-up — the Tab5 can only claim the card during a short window early in boot.

When the mirror runs

The microSD card and the WiFi co-processor share a bus on this hardware and cannot both use it reliably. Rather than fail at an unpredictable moment, the Tab5 picks the behaviour that works:

	What happens	SD dot
WiFi off (POTA/SOTA)	Continuous mirroring the whole time the card is in	**Green**
WiFi on	One complete backup within a few seconds of switching on, then mirroring stops	**Yellow**

Either way your QSO log, config, and LoTW certificate and key are backed up. With WiFi on, QSOs made later in that session reach the card at the **next start-up** — so if you have been operating with WiFi up and want them on the card now, restart the Tab5.

If no card is inserted the dot is absent, which is not an error.

> ⚠️ **The card holds credentials.** A full backup that can *restore* a station necessarily includes secrets: `qmx-config.txt` stores your WiFi password and QRZ/eQSL logins in clear text, and `lotw_key.b64` is your LoTW **private key**. Keep the card as physically secure as a house key. (The on-card `README.txt` repeats this warning.)

> The diagnostic log is always-on regardless of whether an SD card is present. If no card is inserted, the log still persists to internal flash (see Diagnostic Logging above) and survives a power-off.

8.16. Activation (POTA / SOTA)

When you are activating a park or a summit, every contact you make has to be logged as belonging to **that reference** — it is what POTA and SOTA read to credit the activation, for you and for the people who worked you.

Open **Activation (POTA/SOTA)** in the settings drawer, pick the scheme, type the reference, and tap **Start activation**. From then on every logged QSO carries it automatically. There is nothing to remember per contact.

The screen shows your **contact count against the threshold** — ten for POTA, four for SOTA — and turns green when you reach it. That number is read from the log itself, so it survives a reboot and can never disagree with what was actually written.

Stopping matters as much as starting. While a session is running the main button is a red **Stop activation**, and the drawer button itself shows the live reference rather than a generic label. The failure that actually happens is driving home with it still switched on, quietly stamping every later contact with a park you have left.

The session persists across a restart on purpose — a battery change in the field should not lose it — but it is deliberately **not** included in a config backup, so restoring an old backup can never re-activate a park you are nowhere near.

Chasing works without any setup: if you work a station the panadapter has seen spotted at a park or summit, that reference is written into your log entry automatically.

Uploading just one activation: the web UI's ADIF download can be limited to a single reference, so you get that park's log rather than your entire file.

8.17. SWR protection

While transmitting, the QMX reports its SWR back over the control link. If the reading reaches the limit you set here the transmitter is **latched off** — nothing will transmit again until you clear it by tapping the red warning on the FT8 screen.

Choose **Off, 2.0:1, 2.5:1, 3.0:1 or 4.0:1**. The default is **3.0:1**.

In FT8 the burst is cut short; the SWR is sampled part-way through and the transmission stops there. **In FT4 the check happens after the burst**, so the one burst completes and every later one is blocked. That is not an oversight: FT4 symbols are 48 ms and a control-link query takes about 50 ms, so asking mid-burst would disturb the transmission it is trying to protect. A reading is only acted on if there is real power behind it — the radio can report a meaningless ratio when the amplifier is not actually loaded. That is high enough not to trip on a merely mediocre match, and low enough to stop a burst going into a disconnected antenna, a wrong-band antenna or a damaged feedline — which is how this usually goes wrong in the field. An FT8 transmission is nearly thirteen seconds of continuous key-down, so there is real time to save.

The latch is deliberately sticky. A bad antenna does not repair itself, and resuming automatically would simply transmit into the same fault on the next slot.

!!! note "Set it to Off if you use a tuner" With a matched antenna the protection never fires, but if you are deliberately loading something unusual you may prefer no interference at all.

8.18. Propagation feedback — who is hearing me

This asks PSK Reporter which receivers have copied **your** callsign recently. It is the reverse of the reports the panadapter *sends*, and the two are separate settings — you can have either without the other.

It answers a question no amount of local processing can: **am I actually getting out?** The valuable case is the mismatch. Stations you can hear that cannot hear you is a transmit-side problem — antenna, power, a bad connector — and from the receiving side alone it looks exactly like a dead band.

This one lives in the browser, not on the Tab5. There is no drawer control for it — the answer is a list of stations with distances and bearings, which wants a screen you are already sitting in front of. In the web UI open **Settings → Spots & reporting** and tick **Propagation feedback (who is hearing me)**, then **Miscellaneous → Who is hearing me** for the list: receiver, country, distance, bearing and the signal report they gave you, sorted by distance.

It is read-only and asks at most once every five minutes, which is the rate PSK Reporter requests.

!!! note "An empty list is the normal answer if you have not transmitted" Nobody can report hearing you until you have been on the air. Give it a few minutes after a CQ run.

8.19. ADIF & Logging

ADIF Log — View the QSO log on-device. The viewer shows a proper column table:

Column	Content
Call	Callsign
Country	DXCC entity (looked up from the callsign prefix)
Mode	FT8 or FT4
Band	Band (20m, 40m, ...)
Date	UTC date
Time	UTC time
Sent	Your signal report (SNR)
Rcvd	Their signal report (SNR)

A sticky header row stays pinned while you scroll. Even-numbered rows are lightly shaded so long logs stay easy to scan.

Today/All filter — the viewer opens on **Today** (falling back to All when nothing was logged today). The toggle button shows the view you *switch to* by pressing it; the title shows the current view with counts.

POTA activation counter — in the Today view the title reads "Today: N (M total)" and turns **green** once today reaches 10 QSOs — a valid POTA activation.

Delete a single record — **long-press** a QSO row: the row highlights red and list scrolling locks. Drag up/down to move the highlight, then release — a Delete/Cancel bar appears at the bottom. **Delete** removes just that one record (useful for duplicates).

Delete all — the red-bordered button at the bottom-left erases the **whole** log. Two-tap confirm: the first tap arms it (the label changes to "ALL *N*?"), a second tap within 5 seconds deletes; wait and it disarms itself. There is no undo — download the ADIF from the web UI first if you want a copy. Handy before a POTA activation: start with an empty log and the ADIF at the end is exactly the file you submit.

Use the web UI to download the full ADIF file for import into WSJT-X, EQSL, or any other logging software — or view and edit it in the browser (**QSO Logs** → **View / edit log**).

Exclude Worked Before — When FT8 filtering, skip stations you've already logged QSOs with (requires you to import your own prior ADIF log first).

8.20. Config Import/Export

Config Download — Export all settings + memory channels + ADIF log as a text file (INI format).

Config Upload — Restore settings from a backup file (settings only; memory channels merge).

Use this to:

- Back up your settings before a factory reset
- Transfer settings to another Tab5
- Recover from accidental changes

8.21. Advanced / Expert

These two live in the browser, not in the drawer — a reset you can trigger by mistake on a touchscreen in the field is a worse idea than one that needs a computer. Both are in the web UI's **Miscellaneous ▲** menu.

Reset settings — Erases the stored settings and memory channels, returning everything to defaults, and reboots. Use if something is stuck in a state you cannot get out of. Take a **Config** ↓ backup first and you can restore it in seconds.

Reset WiFi — Erases the WiFi credentials and the stored radio/link state. This is the one for a WiFi setup that will not come up no matter what you enter.

Neither of them touches your QSO log. The ADIF log lives in a separate area of flash that is deliberately left alone, so a reset cannot cost you contacts — and neither can a firmware update, which is why a normal flash keeps your log. The only things that erase the log are the **Delete all** button in the ADIF viewer, the same in the web log viewer, and a **clean/erase flash** from the flasher.

Next: Troubleshoot an issue via Troubleshooting or explore the API.

The settings drawer, opened by swiping in from the right edge. It is grouped — **Station, Device, Radio, Network, Display, FT8, Spectrum** — with a **Basic / Expert** toggle that keeps the list short until you need it to be long. Changes save themselves; there is no OK button to hunt for.

📖 Every control, group by group: tab5.lav.dk/guide/settings

---

9. Reference

Gestures

Gesture	Where	Effect
Tap	Spectrum / waterfall	Tune to tapped frequency (snapped)
Touch + drag (hold ~250 ms first)	Spectrum / waterfall	Cyan cursor snaps grid-to-grid; tunes on release
Fast one-finger horizontal swipe	Spectrum / waterfall, any zoom	Pan/"stroll" the view; retunes to the new centre on release
Double-tap	Spectrum / waterfall	Reset zoom and pan to ×1.0 / centred
Pinch (two fingers)	Spectrum / waterfall	Zoom ×1.0–×24.0
Two-finger drag	Spectrum / waterfall (zoomed)	Pan the zoomed window
Swipe → from left edge	Left edge strip	Toggle Panadapter ↔ FT8 screen
Swipe ← from right edge	Right edge strip	Open settings drawer
Tap right grip handle	Right edge	Open settings drawer (alternative)
Swipe →	Open drawer, or spectrum while drawer open	Close settings drawer
Swipe ↑ from bottom edge	Bottom edge strip	Open memory channel picker
Tap or swipe ↓	Top-bar item (Band/Mode/BW/Freq/Zoom)	Open that item's selector

Gesture	Where	Effect
Touch and hold ~250 ms	FT8 decode list row	Dim preview at ~80 ms; full selection at 250 ms (scroll locks, drag moves highlight, lift to confirm)
Quick swipe	FT8 decode list	Scroll the list normally

Per-unit IF calibration

The QMX's +12 kHz IF injection varies slightly between units. If signals appear consistently shifted left or right of where the QMX is actually tuned, open the settings drawer → **IF calibration** slider (±100 Hz, persisted to NVS). Default 0. Reported by Ken KF0AYY, whose unit needed about −55 Hz to centre correctly.

Hardware

- **M5Stack Tab5** — ESP32-P4 v1.3 (ECO2), ST7121 or ST7123 5" 720×1280 MIPI-DSI touch display, 32 MB PSRAM, ESP32-C6 co-processor for WiFi
- **QRP Labs QMX or QMX+** — firmware 1.03.002 or newer (the 1.04 betas also work, and unlock AM mode + Antenna Tune)
- **USB-A → USB-C data cable** between Tab5 USB-A host port and QMX (full data, not charge-only)
- **USB-C power supply** for the Tab5 (5 V / 2 A or better, or internal battery)

```
┌─────────────┐   USB-A → USB-C (data cable)    ┌─────────────┐
│  M5Stack    │  ──────────────────────────────►│   QMX+      │
│  Tab5       │   UAC: I/Q audio (48 kHz stereo)│             │
│             │   CDC-ACM: CAT control          └─────────────┘
│             │
│             │   USB-C (any power cable)       ┌─────────────┐
│             │  ──────────────────────────────►│  5 V source │
└─────────────┘                                 └─────────────┘
```

What works without the QMX connected

Works without QMX	Needs QMX connected
UI, settings drawer, gestures	Spectrum + waterfall (no I/Q = flat line)
WiFi, web UI, screenshots	Top bar Band / Mode / BW
Callsign / grid / WiFi credentials	Tap-to-tune, mode popup (CAT writes)
Brightness, colour map, dB range	S-meter, FT8 decode/TX, QMX firmware readout

If the spectrum is flat and the top bar shows `---`, that is almost always the radio not being seen over USB — which loops back to the cable (or the QMX being off / in flash mode).

Spectrum signal shifted / mirrored, tunable across the whole 48 kHz window

This means the QMX never confirmed IQ mode for the session — without it the radio streams plain (non-IQ) audio instead of a centred baseband, so the signal appears at the wrong place and slides across the full 48 kHz window as you tune. As of v0.19.3 the Tab5 retries the IQ-mode handshake automatically at connect (up to 4 attempts) and this resolves itself almost every time; if it still fails after all retries, a red banner appears across the top of the screen telling you so immediately, instead of leaving you to figure it out from a shifted waterfall. If you see the banner, power-cycle the QMX (forces a fresh handshake on reconnect) or check the QMX's own **System Config → IQ Mode** setting.

Reporting hardware issues

Near the top of the boot log you will see:

```
I (xxxx) bsp_info: === TAB5 BSP INFO ===
I (xxxx) bsp_info: chip:     ESP32-P4 rev v1.3
I (xxxx) bsp_info: psram:    30 MB
I (xxxx) bsp_info: panel:    ST7123 (inferred from touch)
I (xxxx) bsp_info: touch:    ST7123 @ 0x55
I (xxxx) bsp_info: heap:     230.5 kB internal free, 28.80 MB PSRAM free
I (xxxx) bsp_info: idf:      v5.4.4
I (xxxx) bsp_info: firmware: v0.21.0
I (xxxx) bsp_info: =====================
```

When opening a hardware issue, paste this block — the `panel` and `touch` lines are the first thing needed to identify which Tab5 hardware revision you have.

> **Note on `reset_reason`:** a deliberate reset-button force-off returns `panic/exception`, not `external-pin/power-on`. A genuine firmware crash always prints a `Guru Meditation` register + backtrace dump immediately *before* the reboot. No backtrace = abrupt power-off, not a crash.

Appendix — Technical Reference

The sections below are the remaining, more technical parts of the same project README (build instructions, internal design notes, roadmap) — included here so the links above resolve, rather than dead-ending. Skip ahead if you only want operating instructions.

Appendix A — Build from source

Requires **ESP-IDF v5.4.4** — pinned; do not upgrade. ESP32-P4 v1.3 silicon requires `CONFIG_ESP32P4_REV_MIN_0=y` and CPU capped at 360 MHz, both baked into `sdkconfig`.

```
# Activate the IDF environment, then:
idf.py build flash monitor
```

Or with the `qmx` PowerShell helper — add to your `$PROFILE` and adjust the COM port:

```
function qmx {
    param([string]$cmd = "fm")
    if (-not $env:IDF_PATH) { & C:\esp\v5.4.4\esp-idf\export.ps1 }
    switch ($cmd) {
        "b"   { idf.py build }
        "f"   { idf.py flash }
        "m"   { python -m esp_idf_monitor -p COM3 build\qmx_panadapter.elf }
        "fm"  { idf.py flash; python -m esp_idf_monitor -p COM3
         build\qmx_panadapter.elf }
        "bfm" { idf.py build flash; python -m esp_idf_monitor -p COM3
         build\qmx_panadapter.elf }
    }
}
```

Exit monitor: `Ctrl+T` then `Ctrl+X`.

Appendix B — Under the hood

I/Q balance correction

The QMX presents I/Q audio with a small but measurable amplitude imbalance and quadrature error between channels. Without correction, every signal produces a mirror image across the centre frequency (~30 dB weaker), cluttering the waterfall on a busy band.

A blind adaptive Gram-Schmidt orthogonaliser runs sample-by-sample in the audio pipeline before the FFT. It maintains running estimates of:

- **DC offset** on each channel (τ = 1 s)
- **Power** in I and Q (τ = 200 ms)
- **Cross-product** I·Q (τ = 1 s) — non-zero cross-product means the channels are not orthogonal

Correction per sample:

```
i_out = i
q_out = (q − K_phi · i) × K_amp
```

No calibration step needed; the estimator converges on ambient band noise within a few seconds of any signal. A two-speed startup runs all alphas at 8× for the first 2 s of real signal after each reset (effective convergence ~125 ms), then drops to steady-state. Toggle in the settings drawer; re-enabling resets the estimator so it reconverges from a clean state.

Quirks and trade-offs

PI4IO expander must be initialised before display bring-up. The Tab5's PI4IO I/O expander holds `LCD_RST` and `TP_RST` low at chip power-on. Without explicit init, on a true cold boot the panel never comes out of reset and the DSI FIFO hangs. Soft resets mask this because the expander retains state across ESP32 resets. `display_init` calls `bsp_i2c_init() + bsp_io_expander_pi4ioe_init()` first, then waits 120 ms.

Patched components in `components/`. `espressif__usb_host_uac/` has `create_background_task = true` — required for UAC and CDC-ACM to coexist on the same USB host. `espressif__esp_lcd_touch_st7123/` makes `FW_VERSION_REG` the only mandatory register read (ST7121 NACKs the others and also adds a `max_touches > 10` bounds clamp). Do not replace either with the registry version without re-applying the patches.

LVGL software rotation (~50% FPS cost). The panel is natively portrait; landscape is achieved via `lv_display_set_rotation(LV_DISPLAY_ROTATION_90)`. Every flush goes through `rotate90_rgb565`. ~13 fps landscape vs ~22 fps portrait — acceptable for a panadapter. PPA hardware rotation conflicts with the USB host stack over DW-GDMA channels and silently kills UAC + CDC-ACM; do not set `CONFIG_LVGL_PORT_ENABLE_PPA=y`. A full native-portrait rewrite would recover the lost FPS but isn't planned — accepted as a permanent trade-off.

IDLE watchdog disabled. The LVGL rotation pipeline keeps CPU0 busy past the default watchdog window. `CONFIG_ESP_TASK_WDT_CHECK_IDLE_TASK_CPU0/CPU1` are off; the app-task watchdog (30 s) is still active.

Cross-thread CAT writes must go through the poll task. Writing CAT commands directly from the LVGL thread races the FA/MD/FW poll on the same CDC pipe — commands interleave and the QMX returns `?;`. Pattern: stash the request in a `volatile` and let `poll_task` drain it on its next cycle. See `s_pending_ssb_bw` / `cat_request_ssb_bandwidth()` in `cat.c`.

SSB filter bandwidth needs three coordinated writes. `MMSSB|Filter RX=<hz>;` commits the value (persists, shows in QMX menu); `MMSSB|Bandwidth=<hz>;` applies it live; `FW;` polling must be suspended while the width is pinned because reading the filter makes the QMX revert it. Dead ends: `FW<nnnn>;` CAT set returns `?;`; re-asserting the same mode digit does not reload the filter; `Bandwidth` write alone reverts on the next poll.

HTTPS needs mbedtls allocating from PSRAM, not internal RAM. The QRZ/eQSL upload features (v0.16.2) are this firmware's first-ever outbound HTTPS connections — SNTP, the only prior network client, is UDP. With the default `CONFIG_MBEDTLS_MEM_ALLOC_MODE=MBEDTLS_INTERNAL_MEM_ALLOC`, the TLS handshake's 16 KB+ buffers competed for the ~200 KB internal DRAM already under pressure from USB host/audio/FFT/LVGL and every connection failed with `ESP_ERR_HTTP_CONNECT`. Fixed by switching to `MBEDTLS_EXTERNAL_MEM_ALLOC` in `sdkconfig`, which moves those buffers into the 28 MB of free PSRAM instead.

Project layout

```
main/
  main.c                    app_main, task launch, orchestration
  display/display.c         BSP bring-up (PI4IO + LCD + touch)
  ui/ui.c                   LVGL widgets, touch handler, canvases
  ui/ft8_screen_view.c      FT8 decode list, touch-drag row selection
  ui/ft8_tx_modal.c         TX confirmation modal
  cat/cat.c                 USB CDC-ACM + Kenwood CAT
  audio/audio.c             USB UAC + ring buffer producer
  dsp/dsp.c                 FFT, spectrum mutex, DC blocker
  dsp/iq_balance.c          Gram-Schmidt I/Q correction
  ft8_tx.c                  FT8 TX engine (build/arm/run/abort)
  ft8_test.c                FT8 slot loop (RX decode / TX burst)
  ft8_qso.c                 Auto QSO state machine
  ft8_sim.c                 FT8 simulation mode: phantom-station practice QSOs
  ft8_status.c              Mutex-protected FT8 status string
  adif/adif_log.c           ADIF QSO logging
  adif/qrz_upload.c         QRZ Logbook API upload (one record per request)
  adif/eqsl_upload.c        eQSL.cc upload (batched, multiple records per request)
  rtc/rtc.c                 RX8130CE supercap RTC driver
  time_sync/time_sync.c     Time sync orchestrator
  render/render.c           30 Hz render task, EMA smoothing
  render/render_waterfall.c Waterfall scroll (double-height canvas)
  screenshot/screenshot.c   RGB565 framebuffer capture for /ss.bmp
  storage/settings.c        NVS-backed settings with debounced flush
  wifi/wifi.c               C6 co-processor WiFi + SNTP
  net/webserver.c           HTTP server + WebSocket
  util/fps.c                FPS counter
  util/diag_log.c           Diagnostic log ring buffer (vprintf hook)
```

Appendix C — Roadmap

The full per-version changelog — every release from v0.1.0 onward — lives in **docs/version-history.md**. This section tracks only what's planned next.

Next up

v1.8.1 is here — a fixes release on top of v1.8.0: tap-to-tune across the whole spectrum, CW centre matching the radio, the CW transmit offset tidying up VFO B, RF gain and volume agreeing across both screens, browser spot labels that no longer steal clicks, settable seconds for POTA use with no WiFi, and an option to hide the RIT button. Previously in **v1.8.0 — RIT you set by tapping, summit spots, and a browser that matches the Tab5.** Arm RIT and tap a caller to receive off your transmit frequency; SOTA summit activations join POTA, RBN and the DX cluster on the spectrum; Fox/Hound works DXpeditions from the hound side; and the browser gained RIT, activation start/stop, the last Tab5-only settings and a spectrum and waterfall drawn the way the Tab5 draws them. Next on the bench:

- **Web-UI audio streaming.** Listen to the receiver in any browser on your LAN — demodulated on the Tab5, no PC. Already working in development; held back for quality tuning and an overnight streaming soak. Server mode (screen off, device just serves) rides along.
- **CW page.** Canned-message CW TX memories first; decoded-CW display after (the QMX decodes internally — mirroring it over CAT looks cheap).
- **Binaural CW audio.** Asked for by Roy KI0ER, and shaped by Don N2VGU and Michael KZ4LY: a stereo sound stage so two stations a few tens of hertz apart land in different places in your head, with the **stage width a setting** rather than a fixed angle. The DSP is small — the Tab5 already receives I and Q separately — but it needs the Tab5's own audio output path, which is the same rework the CW page waits on.
- **Tab5 audio output rework.** The blocker under both of the above: the output task must not run at all in FT8/FT4 (Michael KZ4LY's suggestion), since merely existing at a higher priority than the FFT consumer cost decode yield.

Longer term

- **CW decoder.** Goertzel-based, text scrolling under the spectrum. The QMX already does this internally — question is whether to mirror its output via CAT or run a parallel decoder on the Tab5.
- **Tab5 speaker / headphone audio.** Demodulated CW/SSB passband audio out of the Tab5's own jack, so the operator can monitor without the QMX's audio path.
- **Extended waterfall history.** PSRAM has room for several minutes of scrollback; two-finger drag to scrub through history.
- **QMX (small) support.** Same UI, different USB endpoint config and band table.
- **JS8 / RTTY modes.** See `docs/js8-feasibility.md` and `docs/rtty-feasibility.md`.
- **DSP polish.** Noise reduction, auto-notch.

Appendix D — Related projects

- DX-FT8 by Barb (WB2CBA) — open-hardware FT8 tablet transceiver; an inspiring reference for a similar use-case
- `qrp_companion` by Zhenxing Han (N6HAN) — Tab5 companion for QMX with audio + CAT; source of the polling audio task pattern and battery readout approach
- `ft8_lib` by Karlis Goba — FT8 encoder/decoder vendored as `components/ft8_lib`

Glossary

Common terms and acronyms used in this guide:

Term	Meaning
CAT	Computer-Aided Transceiver — radio control protocol (Kenwood-style commands via serial/USB)
CDC-ACM	Communications Device Class / Abstract Control Model — USB standard for serial ports
CQ	General call to any station (not directed at anyone specific)
CW	Continuous Wave — Morse code mode
DSP	Digital Signal Processing — mathematical signal analysis and filtering
FFT	Fast Fourier Transform — algorithm to convert time-domain audio into frequency spectrum
FT8 / FT4	Digital modes for weak-signal HF communication (15-second vs 7.5-second slots). Both fully supported (FT4 re-enabled in v0.21.0).
GPIO	General-Purpose Input/Output — microcontroller pins for digital signals
I2C / SPI	Serial communication protocols for connecting peripherals (sensors, displays, etc.)
IQ	In-phase / Quadrature — stereo representation of RF signals (real + imaginary parts)
LVGL	Light and Versatile Graphics Library — open-source embedded UI toolkit used for the display
NVS	Non-Volatile Storage — persistent memory on the ESP32 (survives power cycles)
PSRAM	Pseudo-SRAM — extra RAM on the Tab5 (used for large buffers like waterfall history)
QMX / QMX+	QRP Labs HF transceiver — the radio this panadapter controls and receives audio from
QSO	Radio contact / conversation between two stations

Term	Meaning
RTC	Real-Time Clock — battery-backed timer on the Tab5 (keeps time during power-off)
SNTP	Simple Network Time Protocol — synchronizes system clock via WiFi/internet
SWR	Standing Wave Ratio — antenna impedance matching metric (1.0 = perfect)
TX / RX	Transmit / Receive — keying the radio and listening
UAC	USB Audio Class — standard for streaming audio over USB
USB	Universal Serial Bus — physical connector and protocol (carries both audio and CAT commands)
UTC	Coordinated Universal Time — timezone-independent time standard for FT8 slot alignment
VFO	Variable Frequency Oscillator — the radio's tuning dial / frequency setting

---

Contributing

This is a solo project — all coding is done by the author (OZ1LAV) at his own pace, and **pull requests are not being accepted right now**. Bug reports, feature requests, and field reports are very welcome, though, via GitHub Issues or the QRPLabs Groups.io thread. See CONTRIBUTING.md for details.

Appendix E — License